

The Globalization Origins of Autocratic Rise: Engaged Reformers and the Post-Cold War Divergence

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Abstract

Conventional wisdom holds that globalization promotes political liberalization and that inclusive institutions favor growth. Yet many autocracies consolidated and thrived economically after 1990, especially in trade. This article argues that globalization alters the payoff structure of development in a way that advantages a specific authoritarian rule—economically engaged and reformed autocracies—in external-demand competition, despite their weak internal demand. Autocratic institutional features conventionally viewed as liabilities in closed economies, such as centralized power or weak accountability, can become advantageous under engagement-and-reform configurations. Specifically, trade integration and domestic institutional reform jointly supply credibility for global market participation, upon which autocratic institutions are less costly for amplifying export competitiveness. This globalization-contingent, demand-competing advantage substitutes for politically inclusive institutions. Empirically, I connect macro-level patterns to micro-level mechanisms, showing how autocratic “engaged reformers”—accounting for over 90 percent of autocracies’ output and extending beyond China and oil states—outperformed others in exports, among broader economic outcomes. The findings illustrate how globalization, interacting with domestic regimes, alters the institutional logic of competitive advantage, while revealing the limits of authoritarianism alone.

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“... the view that democratic institutions are at best irrelevant and at worst a hindrance for economic growth has become increasingly popular in both academia and policy discourse ...”

– Acemoglu et al. (2019)

1 Introduction

Economic globalization deepened markedly after the Cold War (Baldwin 2016; Rodrik 2012), along with the widely shared expectation that economic liberalism would advance political liberalization (Fukuyama 1989; Ikenberry 2001). The post-1990 era has instead produced democratic backsliding some attribute to globalization (Autor et al. 2020; Inglehart and Norris 2017; Rodrik 2017).¹ More striking is what I term “autocratic rise”—many autocracies reconsolidated and thrived (Diamond 2015; Ekiert and Dasanaike 2024), and outpaced the economic performance of democracies (see Figure 1 and Section 2).²

Why have autocracies risen economically alongside globalization? Studies of globalization and democracy/democratization (e.g., Autor et al. 2020; Milner and Mukherjee 2009; Rodrik 2012) offer limited answers. The extensive literature on regime type and economic performance provides notably mixed conclusions (e.g., Acemoglu et al. 2005; Barro 1996; Chandra and Rudra 2015; Przeworski et al. 2000) or no comparison (Acemoglu et al. 2019); importantly, this literature, well suited to an earlier period, emphasizes domestic factors and largely brackets globalization.³

This paper places the contextual role of globalization at the center. I argue that autocratic rise is driven by a specific form of autocratic regime: those that are *both* economically engaged and reformed, or *engaged reformers*—that is, integrated into the global trade regime and having implemented sufficient market-oriented institutional reforms. This two-part thresh-

¹Scholars have called for “defining research programs” to understand this phenomenon (e.g., Drezner 2022).

²Autocracies’ data may be distorted due to currency devaluation (Steinberg and Malhotra 2014) or under-reporting (Martinez 2022). Trade data, reported by others, are more accurate.

³This literature relies predominantly on pre-2000 data.

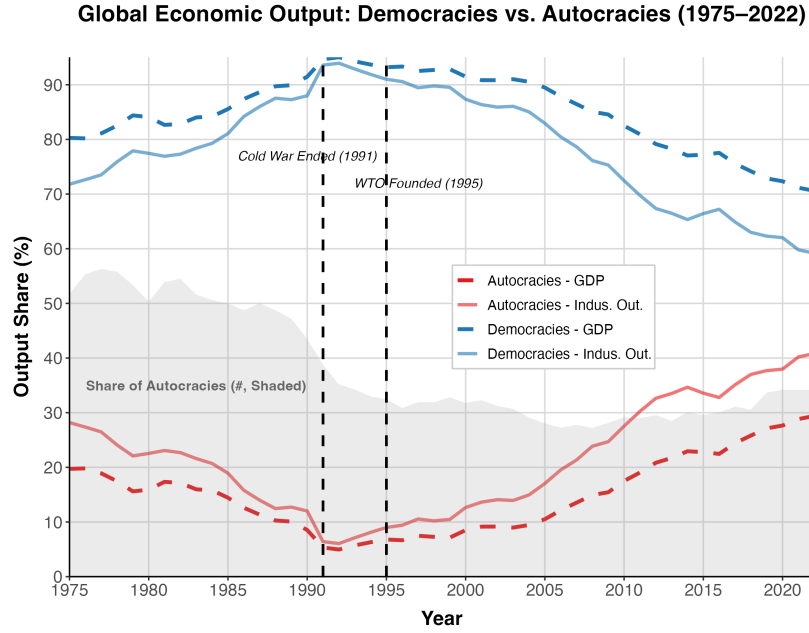


Figure 1: The Distribution of Production Between Democracies and Autocracies (World Bank). *Note:* I use a conservative measure for Autocracy (Freedom House (FH) Index ≥ 10), with similar patterns for using Polity or excluding China (Appendix A.6). In 2020, China accounts for 62% of autocracies’ GDP. Shaded area depicts the count proportion of autocracies, indicating the stalled democratization.

old is the theory’s *scope condition*: autocracies that fulfill it under globalization gain trade advantages over their democratic “counterfactual” in competing for global demand (e.g., consumption markets), despite weak domestic demand (Acemoglu and Robinson 2012; Bueno de Mesquita et al. 2003; see also Figure A.3). The Cold War era featured constrained economic exchange and regional blocs. Most developing countries—authoritarian or democratic—were inward-oriented, practicing planned economies or import substitution industrialization (ISI), maintaining “trade skepticism,” and eventually confronting stagnation (Sachs and Warner 1995).⁴ With globalization, market-oriented, outward-oriented reforms became the prevailing development template (Quinn and Toyoda 2007; Simmons and Elkins 2004). Importantly, many were engaged by Western-led institutions such as the WTO and IMF, which bolstered their credibility, reform progress, and access to markets, capital, technology, production networks, and policy guidance (Antràs 2015; Arias et al. 2018; Baccini and Urpelainen 2015; Baldwin 2016). This is essential for autocracies, whose political non-inclusiveness undermines

⁴With some exceptions, e.g., export-oriented Asian countries, which align closely with the theory.

commitment credibility and domestic demand.

The notion of “autocratic advantage” is not new (e.g., Wade 1990; Weede 1996). Recent research documents, for example, autocracies’ propensity to sign international agreements (Arias et al. 2018) and economic resilience (Lipsky 2018). Yet the mechanisms underlying the post-1990 autocratic rise remain underspecified. I argue that globalization, as a macroeconomic-institutional shift, alters the institutional logic of competitive advantage. In inward-oriented economies, growth relies on inclusive institutions to mobilize domestic entrepreneurship and sustain domestic demand. Autocratic institutional features—centralized authority or low accountability—are costly: they weaken credibility, suppress consumption, and invite predation. Globalization expands economic exchange and diffuses outward-oriented reform; more importantly, it shifts the payoff structure of development by rewarding outward- rather than inward-oriented development models. Once engaged and reformed, (outward-oriented) autocracies possess necessary credibility for investors and global market participation. On top of this, the same autocratic features—still consumption-suppressing—remain less costly (i.e., facing fewer constraints) for building export competitiveness, through *active* (e.g., selective liberalization, targeted reform, or competitiveness-enhancing policies) and *passive* channels (e.g., weak labor protection, limited redistribution, or restrained distributional politics). Underpriced wages, for instance, only weaken domestic consumption unless the resulting cost competitiveness captures more external markets. Conversely, hand-tying constraints that define democracies—deliberation, veto players—become costlier in demand competition. This globalization-contingent advantage thus substitutes for inclusive institutions, which matter more for domestic demand, reconciling autocratic rise with the conventional wisdom (Acemoglu et al. 2001; North and Weingast 1989).

In short, post-1990 globalization alters the “game” of development by intensifying demand competition and rewarding externally rather than domestically oriented growth models, advantaging a particular institutional configuration—autocratic engaged reformers. Consider several examples that differ sharply from unreformed, isolated autocracies such as Mao’s China and Saddam’s Iraq, or “Autocracy 1.0” (Lind 2025; Yang 2024).⁵ Instead, they emu-

⁵See Table A.1 for more examples.

lated liberal economic institutions and substantially owed their success to external demand: Most post-communist states had reoriented their growth to Western markets by 2000 (Åslund 2012), whereas China or Vietnam heavily depended on exports (Feenstra and Wei 2010).⁶ Though resource-endowed, Qatar and the UAE took off only after 1990, leveraging WTO membership and heavy industrial investments (Hvidt 2013)—alongside dos Santos’ Angola to a lesser extent. Tunisia, under Ben Ali, prioritized export-led manufacturing and was the first country to sign the Euro-Mediterranean Association Agreement, tripling exports before stagnation following democratization.

I test the theory in post-1990 globalization, which profoundly transformed global trade, finance, migration, governance, and norms. Specifically, I focus on two key trade-enhancing transformations that jointly satisfy the scope condition above: trade integration (primarily WTO accession) and institutional reform (e.g., rule of law, property-rights (PR) protection). Because the theory centers on external-demand competition, I focus on exports as its arena, which drive broader economic outcomes such as growth and innovation (Bernard et al. 2018; Grossman and Helpman 1991).⁷ I find that autocratic engaged reformers benefited most from WTO accession and outperformed other regime categories in aggregate exports, robust to various identification strategies and competing explanations (e.g., catching-up, commodity booms, importing credibility). The same regimes also outperform in associated channels such as foreign investment, savings, and fixed investment.

I probe the micro-mechanisms in two complementary layers. The first is systematic: sectoral patterns show that gains concentrate in the sectors where regime characteristics should be most consequential, and rules against single explanations such as labor-cost-only. The second goes deep through two matched-pair comparisons: evidence from Japanese investor perceptions in Vietnam and the Philippines indicates that the mechanisms are legible to private actors: Japanese firms shifted manufacturing to Vietnam at three times the Philippine rate owing to institutional differences. A close comparison of China and India, whose pre-reform

⁶Today, China produces roughly 30 percent of global manufacturing output yet accounts for about 12 percent of global consumption (World Bank 2024).

⁷Long-run export growth (1992-2015) is strongly correlated with economic growth ($r = 0.74$, author’s calculations). In Appendix, I also show that autocracies run higher external balances that are linked to global demand redistribution (Chinn and Ito 2021).

structural traits and export trajectories matched before diverging, traces the mechanisms to specific policymaking across differentiated regime settings.

The theory, therefore, moves beyond accounts of strong leadership, state capacity, resource endowment, domestic reform alone, inherent authoritarian advantage, or simply Chinese exceptionalism (though it well explains China’s rise). High-capacity socialist countries (e.g., North Korea and Uzbekistan) or resource-endowed states (e.g., Congo and Venezuela) that fail to meet the scope condition underperform. By identifying how autocratic advantage is rooted in the interaction between domestic institutions (still consequential) and globalization, this paper contributes to the long-standing debate over regime type, institutions, and performance (Acemoglu et al. 2001; Przeworski et al. 2000). The same theory systematizes earlier cases such as Korea, Taiwan, or Singapore (e.g., the “developmental state” literature, see Evans 1995) as instances of a broader mechanism combining domestic institutions and the international environment, while explaining why many Latin American autocracies underperformed.⁸

The findings carry implications for contemporary politics. Economic disadvantage erodes democracy’s domestic foundations, feeds authoritarian tendencies, and weakens leverages abroad (Haggard and Kaufman 1995; Levitsky and Way 2006; Przeworski et al. 2000). As citizens trade democracy for performance amid democratic backsliding (Graham and Svolik 2020; Krishnarajan 2023; Neundorf et al. 2024), why democracy underperforms demands explanation (Drezner 2022; Przeworski 2019). Strengthened autocracies, in turn, reinforce a global authoritarian drift (Wright, Frantz, and Geddes 2013; Baturo and Tolstrup 2024), with implications for international security. This paper locates one source at the level of the international economic order, whose tensions run, as Stiglitz (2018, X) puts it, with “trade at the center.”

⁸Pre-1990 Latin American autocracies mostly pursued ISI except for Chile, and post-1990 democratization eliminated institutional trade advantage, leaving only unreformed autocracies such as Cuba, Haiti, or Venezuela.

2 Performance Divergence: A Puzzle for Existing Theories

This section documents the empirical patterns of autocratic economic rise that are difficult to reconcile with existing literature. Figure 2 reports several economic measures closely linked to globalization: mean merchandise exports (% of GDP), trade balance (% of GDP), industrial output (% of GDP), and GDP growth rate of both democracies and autocracies (Polity ≤ 0). Since the early 1990s, the average performance of autocracies has diverged from that of democracies. These patterns persist after removing developed countries, resource-oriented countries (Russia and the OPEC states), or China (see Figure A.5).

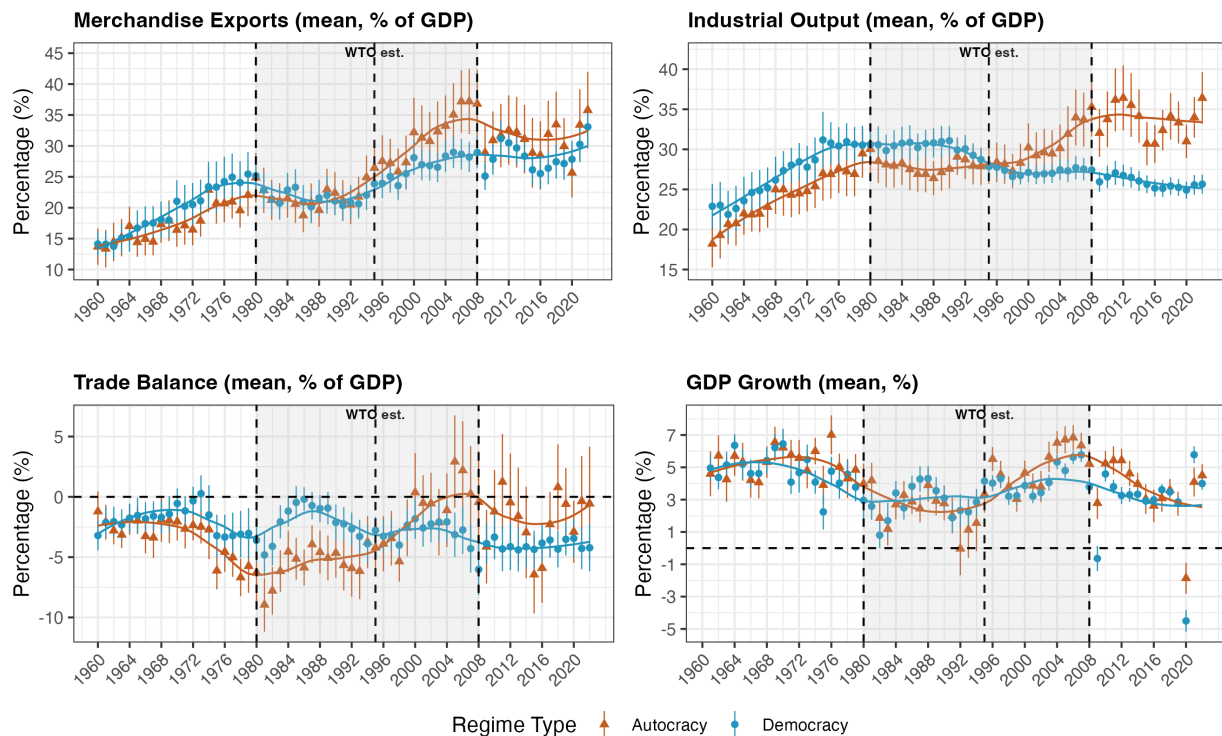


Figure 2: Mean Performance of Economic Indicators between Democracies and Autocracies. *Note:* Bars denote confidence intervals of means. Outliers (top and bottom 2.5%) are removed within each year. 1980 marked roughly the neoliberal turn. The patterns hold after removing developed states, China, Russia, or OPEC states (Figure A.5). Data source: World Bank.

In Figure A.3, between 1990 and 2020, autocracy is correlated with stronger performance across most major economic indicators—growth, fixed investment, industrial output, savings,

exports, reserves, and trade balance—except for consumption and taxation.⁹ A ten-unit increase in Polity (from -5 to 5) is associated with approximately one percentage point lower growth, eight percentage points lower industrial output, and nine percentage points lower exports.

The performance divergence challenges prevailing expectations about regime type and economic performance. Existing research generally finds weak or mixed effects of regime type on growth (Barro 1996; Przeworski et al. 2000). Democracies are expected to outperform because they protect property rights, foster political stability, invest in human capital, and support innovation (Baum and Lake 2003; Weingast 1995), while autocracies may generate growth by limiting redistribution and immediate consumption demands (Bueno de Mesquita et al. 2003). Recent work finds that autocracy is associated with stronger performance only in the post-2000 period (Narita and Sudo 2021) but offers little explanation. In trade, democracies are generally found to export more and adopt fewer protectionist policies, owing to stronger institutional environments (Antràs 2015; Eichengreen and Leblang 2007; Yu 2010).

A related literature emphasizes inclusive institutions for long-run development (North 1990; Acemoglu et al. 2005), suggesting democracy’s advantage, or within-country transition effect (Acemoglu et al. 2019). Another strand of literature reveals possible “autocratic advantages,” including greater policy decisiveness and flexibility (Becker 2010; Wade 1990; Weede 1996), while recent work distinguishes reformed authoritarian regimes from purely repression-based ones (Lind 2025; Yang 2024).

Existing explanations, largely domestic in focus and relying on pre-2000 data, may well account for earlier patterns of development but offer limited insight into autocratic rise under globalization. The puzzle is sharpest in that even after 1990, democracies on average exhibit higher institutional quality yet underperform. I contribute to the literature by incorporating globalization as an external context that reshapes institutional advantages.

⁹Controlling for GDP per capita and year fixed-effects for similar development-level comparisons. Correlations are robust to excluding China or OPEC states and including continent dummies and natural resource rents.

3 “Autocratic Advantage” Revisited: Theory

The intuition that some authoritarian regimes may enjoy developmental advantages is not new. Nobel laureates Michael Spence and Gary Becker, for example, have observed that rapid growth requires “strong political leadership” (Spence et al. 2008) or “not heeding constraints” (Becker 2010). Historical authoritarian economic successes such as South Korea, Taiwan, and Chile suggest that strong performance depends on three general conditions: market-oriented institutions, expanded market access, and state-directed policies.¹⁰ This section theorizes when this intuition holds.

The central claim is that post-1990 globalization alters the payoff structure of development by rewarding outward- rather than inward-oriented growth models. This gives a subset of autocracies—economically engaged and reformed—an advantage in competing for external demand over *comparable* democracies. I term them “autocratic engaged reformers.” However, autocracy is not inherently superior. In closed or unreformed economies, non-inclusive authoritarian regimes often generate familiar pathologies: weak credibility, extraction, misallocation, and suppressed domestic demand. But once engaged and reformed, some authoritarian features can become export advantages.

I focus on exports because external demand remains the core of the theory, providing a critical driver for sustaining investment, competition, innovation, and economies of scale in ways that imports or domestic demand alone cannot (Grossman and Helpman 1991; Harrison and Rodríguez-Clare 2010; Porter 1998; Atkeson and Burstein 2010; Burstein and Melitz 2013).

For autocracies, insufficient domestic demand is a structural rather than contingent feature, because their political non-inclusiveness—limited redistribution, suppressed labor rights, and a narrow selectorate—naturally weakens domestic consumption (Bueno de Mesquita et al. 2003). Export-oriented rents are therefore not merely an opportunity but a necessity. For example, even modest export slowdowns can generate social instability in China, where

¹⁰These three conditions align with the existing literature: institutionalism (North 1990), trade (Krugman 1980), and development (Evans 1995; Haggard 1990).

consumption is weak (Campante et al. 2023).¹¹

3.1 Globalization and the Changing Payoff Structure

The global economy after 1990 differs from that of the Cold War decades in the scale and integration of trade, finance, and production, often called “hyperglobalization” (Rodrik 2011; also Figure 3). This, I argue, resulted in two shifts that altered the payoff structure of development.

First, globalization fundamentally reoriented the development logic outward. The pre-1990 world featured prevalent inward-oriented strategies (e.g., planned economy, ISI). For most countries, domestic markets are structurally insufficient to support efficient industrial scale, which explains why strategies like ISI often failed (Rodrik 2016). Globalization relaxed this constraint. Rapid integration of trade and global value chains (GVC), alongside the transfer of productive know-how, made external demand a substitute channel at scale. Entry barriers to external markets, capital, and technology fell dramatically, including to the GVCs that dominate post-1990 trade (Baldwin 2016; Bernard et al. 2018). International institutions reinforced this reorientation by guiding governments worldwide (Simmons and Elkins 2004): the “Washington Consensus” prescribes outward-oriented policies such as export promotion, FDI attraction, production integration, and competitive exchange rates (PIIE 2021), while the World Bank recommends both institution building and active policy support.¹² Moreover, globalization subjects governments to external stakeholders—including international investors and institutions—that leaders cannot ignore as they once could in closed economies (Jahn 2006).

Second, globalization afforded states substantial policy latitude to enhance export competitiveness. While free trade is conventionally regarded as mutually beneficial, its assumptions, such as perfect competition and few frictions, often fail to hold (Stiglitz 2018)—partly why trade liberalization is a one-time gain with debatable long-term effects (Garrett 2000).

¹¹Indeed, autocracies are more likely to join international institutions to offset domestic deficiencies (Arias et al. 2018), and their lower institutional baseline, even post-1990 (Figure 5), suggests greater dependence on external demand.

¹²“Global Economic Prospects” Report, World Bank, 2024.

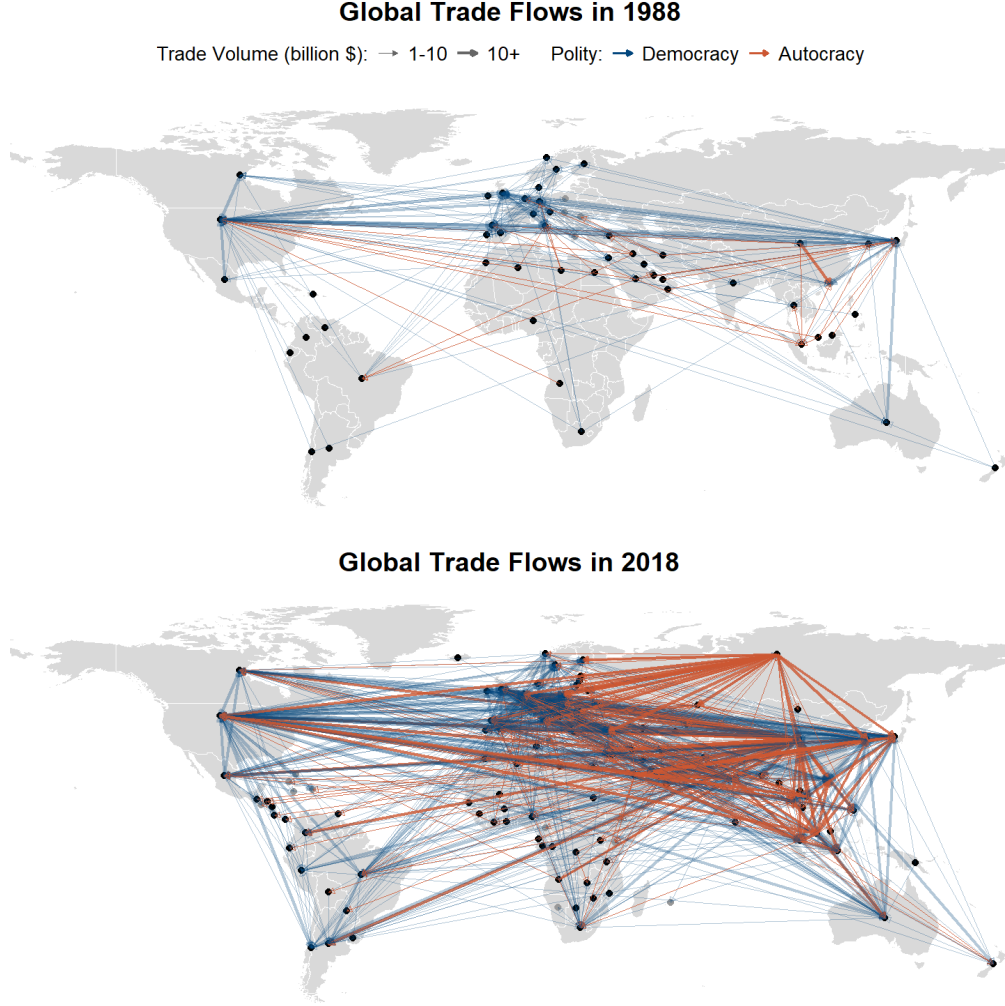


Figure 3: Global Bilateral Trade Flows (1988 vs. 2018). *Note:* For visualization and export-emphasizing purposes, the plot displays only the larger of bidirectional flows colored in exporter's Polity. For example, red lines denote export flows from autocracies. Before 1990, exports were mostly dominated by democracies.

Institutional differences, macro-financial conditions, and policies such as currency devaluation, subsidies, state-directed investments, and wage suppression can all create advantages, especially when technology converges (Antràs 2015; Brander and Spencer 1985; Corsetti et al. 2007).¹³ This is more pronounced when a key assumption of balanced trade, under which such interventions would be neutralized and comparative advantage would materialize, rarely

¹³For example, in the classical Eaton-Kortum model (2002) which assumes a country takes a random productivity draw for goods from a Fréchet distribution (comparative advantages). Currency devaluation ends up acquiring more competitive advantages. Moreover, Arkolakis et al. (2018) show that MNCs choose production location l based on final unit cost: $C_{il} = \frac{\gamma_{il} w_l \tau_l}{z_l}$, where γ_{il} is the foreign production cost, w_l is local wage, τ_l is trade cost, and z_l is firm productivity which can come from within-network technology transfer.

holds.¹⁴ Eaton and Kortum (2002) similarly describe a country’s export competitiveness as “technology adjusted for costs” (also see Bernard et al. 2003; Melitz 2003).¹⁵ Currency undervaluation, for example, leads to production agglomeration (Epifani and Gancia 2017).

As summarized in Table 1, globalization rewards regimes that are less costly for competing for external rents (markets, investment, production, and technology), rather than those relying on domestic markets, which require political inclusiveness. The remainder of this section specifies why that cost is lower for certain regimes.

	Limited Globalization	Globalization
Primary development logic	Domestic demand (inward-oriented)	Demand competition (outward-oriented)
Greater payoff to (institution/policy examples)	Political inclusiveness Consumer welfare Internal redistribution Labor protection Currency appreciation Domestic entrepreneurship	Political autonomy Producer subsidy Export promotion Wage restraint Competitive exchange rate GVC integration/FDI attraction
Primary beneficiaries	Consumers, workers	Producers, exporters

Table 1: How Globalization Alters the Payoff Structure of Development

3.2 Autocratic Advantage under Globalization

The Scope Condition: Engagement and Reform

How, then, does the changed payoff structure advantage certain autocratic institutional weaknesses in exports? The answer begins with a scope condition. Autocracies that preserve state monopolies, insecure property rights, or trade isolation constrain export potential, because

¹⁴E.g., see the discussion of global imbalances in Blanchard and Milesi-Ferretti (2009) and Obstfeld and Rogoff (2009).

¹⁵In Eaton-Kortum model (2002), the proportion of country n ’s total expenditure imported from country i is: $\pi_{ni} = \frac{T_i(w_i d_{ni})^{-\theta}}{\sum T_h(w_h d_{nh})^{-\theta}}$, where T_i represents technology and w_i represents wage or factor cost. Given technology convergence (e.g., due to the diffusion by GVC), wage determines production location. The Melitz model (2003) similarly specifies that firm’s profit ($\pi(\phi) = [(\frac{\phi}{\phi^*})^{\sigma-1} - 1]wf$), which determines firm’s entry into export market, is determined by wage w . Bernard et al. (2003) show that decreased wage increases competitiveness and the range of exports.

they lack external market access and cannot solve the credibility problem for investors. Mao’s China, Nasser’s Egypt, and Nkrumah’s Ghana are examples.

Post-1990 globalization satisfied the scope condition for many: it expanded economic integration, diffused outward-oriented, market-based reform, and revised regimes’ payoff calculus by adding an external source of growth to the internal one. On top of sufficient credibility and trade engagement, some conventionally inferior autocratic features become less costly—that is, facing fewer institutional obstacles—for building export competitiveness. In contrast, inclusive institutions in democracies—political deliberation or distributional accountability—would raise constraints in external-demand competition. Historically, early neoliberal theorists (e.g., Hayek, Mises) viewed democratic politics as constraining economic efficiency (Slobodian 2018), and (semi)authoritarian measures have proven more effective for reform management since Reagan and Thatcher (Bruff 2014).

Note that the engagement-and-reform scope condition is a minimum threshold rather than a determinant of export competitiveness. Many democratic countries with meaningful rule-of-law and WTO membership nevertheless underperform. Selecting into the scope condition reflects a multitude of historical, geopolitical, and domestic reasons more than prior export competitiveness (see Section 4), and the theory compares *within* engaged reformers.¹⁶

From Regime Features to Demand-competing Advantages

Given the scope condition, I identify four autocratic features that can lower the political cost of building export competitiveness. Differing from accounts centered on state-led industrialization, these mechanisms operate through both active (e.g., policy coordination) and passive (e.g., political insulation) channels involving private market players. Both channels are costlier in democracies.

Centralized decision-making power. Autocracies concentrate the capacity to initiate and coordinate policy, as opposed to more fragmented democratic systems. Centralization often leads to expropriation and misallocation, especially in closed or unreformed economies (Acemoglu and Robinson 2012). However, under globalization, it enables reformed autocra-

¹⁶The empirical section deals with potential endogeneity more thoroughly.

cies to compress decision time and pool deployment capacity (Wade 1990; Haggard 1990).¹⁷ Leaders who can, at a lower cost, push for reform, direct concerted policies, sustain consistent planning, and respond swiftly to global market shifts without interagency gridlock hold structural trade advantages (Kohli 2004; Shih 2020). Even relatively centralized Japan and Modi’s regime could implement relatively comprehensive trade-boosting policies, whereas Ukraine’s oligarch-captured, fragmented politics produced underinvestment and external vulnerability (Kuzio 2020).

Weak institutional constraints (ex ante). Autocratic leaders operate under weak constraints from constitutions, legislatures, opposition parties, and norms (Levitsky and Way 2010). Unchecked governments often enable predation in closed economies. In open-and-reform scenarios, however, they can more readily shift resources from other programs towards exports or compromise liberalization commitments with little obstruction. Autocracies’ larger win-sets (Putnam 1988) allow them to conclude international agreements more rapidly and flexibly (Arias et al. 2018). Empirically, autocracies establish more special economic zones offering tax breaks and regulatory relief (Allen and Ge, working paper) and exercise greater control over financial institutions (Brune et al. 2001). In contrast, democracies experience more financial instability, primarily due to their constrained capacity (Lipsky 2018). Within the Gulf region, the UAE’s federal monarchy enabled large-scale industrial and transport investments, while Kuwait’s empowered parliament produced slower policy responses and stalled reform (Hvidt 2013).

Limited societal accountability (ex post). Autocracies remain more insulated from voters and corporatist, labor, or environmental constituencies (Bueno de Mesquita et al. 2003; Rodrik 1999). While non-inclusiveness constrains the domestic economy (Chandra and Rudra 2015), it permits flexible market-favoring policies, even when socially costly (Quinn and Woolley 2001). Regimes insulated from distributional pressures can sustain wage competitiveness, impose high savings for investment, and resist welfare demands—all at the expense of domestic consumption. Bureaucratic promotion in autocracies is upward- rather than voter-oriented (Jensen and Malesky 2018), and many maintain competitive exchange rates

¹⁷It can be economically de-centralized for regional competition (Qian and Weingast 1995).

favoring exporters over consumers (Steinberg and Malhotra 2014). Democracies, by contrast, face pressure to meet immediate demands for redistribution, particularly in poorer countries (Zakaria 1997). Many Latin American democracies maintain labor regulations stricter than those of OECD countries (Feierherd 2024); Brazil, driven by electoral and social pressures, created trade policies largely serving organized interests (Franko 2018).

Mercantilist orientation. While not strictly a regime trait, ideas shape policy preferences. Authoritarian leaders adopted reform out of legitimacy crises, international pressures, and economic necessity (Baturu and Tolstrup 2024; Levitsky and Way 2006; Haggard and Kaufman 2016). They tend to follow a mercantilist and instrumentalist orientation: selectively directing credit, enforcing regulations, and privileging strategic sectors (Tan 2021; Wang 2015). Democracies, by contrast, tend to be more economically liberal and less interventionist (Milner and Kubota 2005). Mercantilism, rather than old-style protectionism, can foster domestic industries or incentivize firms to produce locally, as exemplified in China’s automobile industry and India’s electronics sector under Modi (see case study below).

These mechanisms are not exhaustive: resource endowment creates export foundations and consumption-disregarding norms inherited from socialist economies further suppress domestic demand (Fitzpatrick 1999; Nove 1986). Democracies, for their part, retain genuine advantages in inclusive institutions; yet these advantages are less consequential in global-demand competition.

An inverted-U relationship. Lastly, the mechanisms above share a common foundation in leadership’s discretion, which also implies that institutions, at sufficiently high levels, progressively constrain that discretion by dispersing authority across procedural rules (e.g., institutionalized party consultation, administrative transparency). Thus, autocratic export advantages should be strongest at moderate institutional levels: meaningful reform to supply credibility, but not so much as to eliminate discretion—an inverted-U shape rather than monotonic.¹⁸

¹⁸This is formally illustrated in Proposition 2 (Appendix B.1). There, I formalize the whole argument: facing greater perceived vulnerability and lower political costs, autocracies adopt higher levels of trade-promoting policy. I then extend the classical Eaton-Kortum model (2002) to show how their exports are amplified, conditional on institutional reform and trade integration.

Overall, the coordination capacity and political insulation of autocratic regimes, layered above engagement (market access) and reform (credibility), effectively sustain outward-oriented development under demand competition. The broader implication extends beyond the fact that some autocracies export more; it is that post-1990 globalization altered the institutional logic of competitive advantage, which leads to the unexpected association between autocracy and potentially broader economic strength.

4 Engaged Reformers in Post-1990 Globalization

The theory’s scope condition has two components: trade engagement and domestic institutional reform. This section establishes that post-1990 globalization supplied both, describes the characteristic conduct of engaged reformers, and states the empirical predictions. Other globalization changes are either less relevant to trade (e.g., migration and culture) or predicated on trade integration (e.g., export-oriented FDI).¹⁹

Trade Engagement

Trade engagement or integration serves as the foundation for global production, investments, and technological diffusion (Baldwin 2016). Its principal vehicle is the WTO, which, unlike the already near-universal IMF and World Bank, led the post-1990 expansion of the trade regime, praised as the “most heralded commercial agreement in history” (Goldstein et al. 2007, 38).²⁰ The WTO, previously concentrated in the West, nearly doubled membership from 88 in 1985 to 164 in 2020, extending to autocracies a degree of market access they had not held.²¹ By 2000, most post-communist countries depended on exports to Western markets for growth, especially the most reformist (Åslund 2012). Many non-WTO autocracies enjoyed what I term “semi-engagement”: for example, the Most Favored Nation (MFN) status—a

¹⁹More is discussed in Section 7.

²⁰Regional trade deals usually build on top of WTO principles to address specific issues such as sector-specific trade and dispute resolution.

²¹The 1950s-joiners were mostly advanced democracies such as Germany, Japan, Italy, Austria and Sweden, while the 1960s-joiners were mostly formerly colonies by “automatic accession” under Article 26:5(c). The 1970/80s saw much stagnant accession progress.

WTO principle—from major western countries,²² preferential and regional trade agreements (PTA/RTA), a globalized commodity market, and trade spillovers.

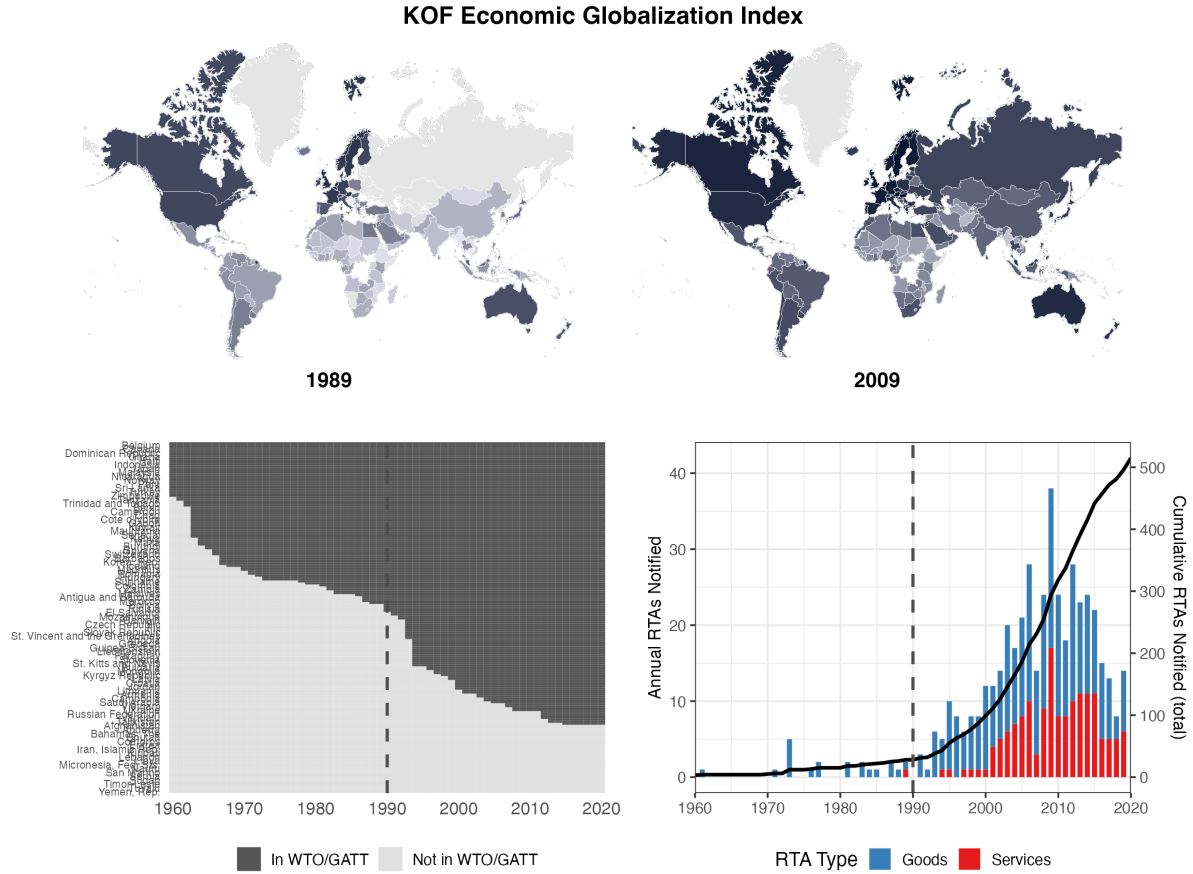


Figure 4: Post-Cold War Economic Globalization Expansion. *Note:* The graphs respectively depict KOF Globalization Index; WTO/GATT Expansion; RTA Proliferation.

The WTO substantially increases trade (Goldstein et al. 2007), usually accruing in private sectors (Baccini et al. 2015). Davis and Wilf (2023) simulate that China and Mexico’s export booms would have occurred earlier had they joined sooner.²³ The WTO, beyond market access, indirectly enhances perceived credibility for politically dissimilar members, which implies that autocracies may benefit disproportionately in a democracy-dominated institution (Arias et al. 2018; Carnegie 2014; Lin et al. 2003). This helps attract FDI and technology (Buthe and Milner 2008) in a trade period dominated by multinational firms

²²See https://1997-2001.state.gov/regions/eap/fs-mfn_treatment_970617.html.

²³Per data, even resource-oriented countries such as the UAE and Oman experienced an immediate export boost upon WTO accession after years of weak exports.

(Bernard et al. 2018). The WTO also guides policies by programs such as the Trade Policy Review Mechanism (TPRM).

Institutional Reform

Under multifaceted pressure, from economic to ideological, during the 1980s, many autocracies (and democracies) followed the “Washington Consensus,” adopting market-oriented reforms bundled with outward-oriented policies as the dominant development template (Quinn and Toyoda 2007).²⁴ Reforms included establishing the rule of law, privatizing state-owned enterprises (SOE), enforcing PR protection, deregulating financial and labor markets, and liberalizing trade and capital accounts (Milner and Mukherjee 2009). Additionally, joining international institutions can reinforce institutional credibility (Arias et al. 2018). Figure 5 shows the historical trends in two common institutional measures: PR protection and rule of law. The former guards investments against expropriation, and the latter governs contract enforcement and dispute settlement (Pandya 2016).

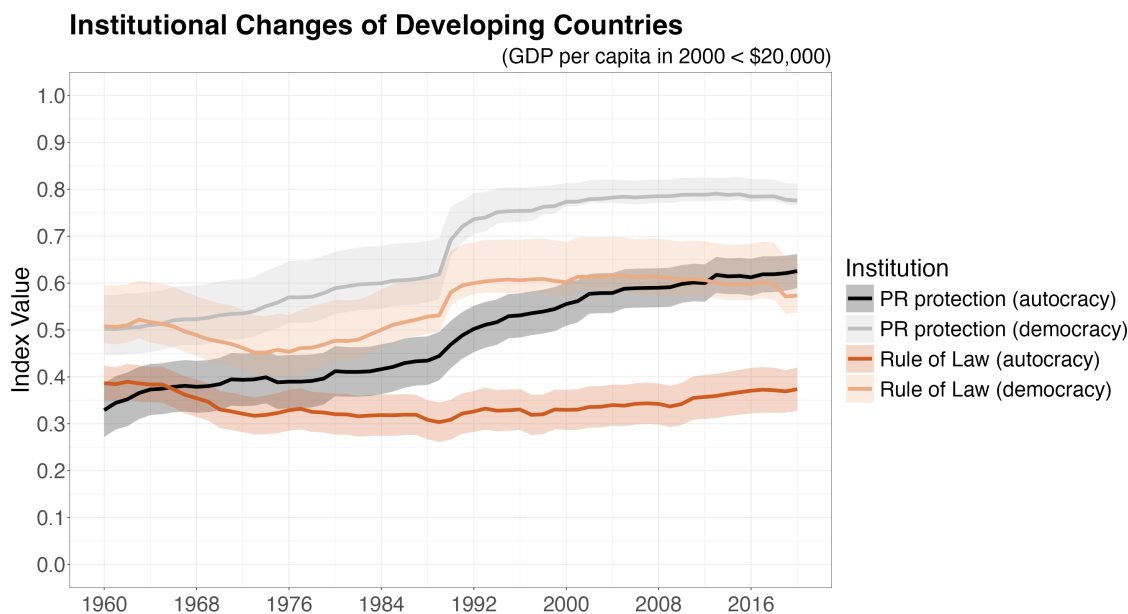


Figure 5: Average Rule of Law and Property Rights Protection. *Note:* Autocracies/democracies are roughly divided by Polity in 1995 (when most countries’ Polity stabilized, see Figure A.7) to ensure temporal country data integrity. Shadows denote 95% CIs.

²⁴Many states established export-promotion agencies—for example, Brazil (ApexBrasil) or Mexico (ProMéxico).

Trade integration without reform yields limited gains (Tang and Wei 2009). Turkmenistan and Azerbaijan entered the 1990s comparable in resources, regime type, geography, and income, and both were semi-engaged through most-favored-nation status.²⁵ Yet Azerbaijan’s substantially higher property-rights protection—0.66 against 0.16—accompanied export growth more than twice that of Turkmenistan.

Selective reform. However, while democracies tend to follow the “Washington Consensus” more faithfully (Harvey 2005; Milner and Kubota 2005), autocracies retain discretionary institutional capabilities to reap the gains of globalization while limiting compliance costs, often without corresponding political reform. China, for instance, selectively liberalized to secure foreign investments and market access (Tan 2021; Wang 2015), extending beyond the reasonable flexibility afforded to new WTO entrants. While democracies have consumers demanding cheap imports and courts enforcing WTO commitments, many autocracies tightly manage exchange rates and capital accounts (Steinberg and Malhotra 2014) and practice mercantilism, including directed subsidies, state-led exporters (e.g., Rwanda’s Crystal Ventures, Morocco’s SNI), economic zones (e.g., UAE’s DMCC), among other measures. Autocracies could ignore domestic labor rights, such as Gulf states’ Kafala or China’s Hukou systems; Post-Soviet states strategically devalued currencies, maintained low taxes, limited social transfers, and suppressed wage growth (Åslund 2012). Selective compliance, moreover, cannot be prevented by independent courts, electorate pressure, or international institutions (Wu 2016).

Combining Trade Integration and Reform

Table 2 classifies all existing autocracies along institutional levels and WTO membership.²⁶ As a face validation, many in the category of “engaged reformers,” beyond oil states and China, performed well.²⁷ In contrast, most states in the other three categories underper-

²⁵Turkmenistan is slightly better endowed: 3.8% of world’s natural gas reserves and 0.04% in oil, while Azerbaijan has 0.5% in natural gas but higher (0.42%) in oil (U.S. EIA).

²⁶Based mainly on institutional levels at the bottom 20 percentile among developing countries in 2010. See Appendix for more details.

²⁷Some countries, such as Cameroon, Mauritania and Togo, do not stand out (although they may still perform better than their “democratic counterfactual”). These newly independent colonized countries automatically joined the GATT; many have ostensibly “copied” economic institutions from former colonizers, but with few substantive reforms (Allee and Scalera 2012).

formed. The same resource-rich autocracies—Algeria, Iran, Iraq, and Venezuela—performed worse than engaged reformers such as Qatar, Kuwait, and Morocco. Many democratic engaged reformers underperformed, such as Argentina, South Africa, and Ukraine. Of the 40 fastest-growing countries from 1992 to 2015, 25 are autocracies, which comprise only 25 percent of all countries.²⁸

	In WTO (before 2015)	Not In WTO
Reformed Institution	“Engaged Reformers” Angola (15.2), Bahrain (5.4◇), Cambodia (21.2), Cameroon (2.6), Chad (22.6▲), China (22.9), Congo Rep. (8.6), Djibouti (10.2), Egypt (8.1), Jordan (6.2), Kazakhstan (21.5), Kuwait (9.7◇), Lao (12.7), Mauritania (5.1), Morocco (6.1), Oman (8.5), Qatar (33.6◇), Russia (8.8), Rwanda (12.2), Saudi Arabia (6.8◇), Singapore (5.1◇), Tanzania (10.1▲), Thailand (5.8), Togo (5.2), United Arab Emirates (13.3◇), Uganda (9.5▲), Vietnam (46.2▲) (Former Autocracies: Brazil, Chile, Hong Kong, Indonesia, Malaysia, Mexico, South Korea, Spain, Tunisia ...)	“Unengaged Reformers” Afghanistan (2.9▲), Algeria (5.2), Azerbaijan (23.8▲), Belarus (12.5), Equatorial Guinea (201.2▲), Ethiopia (12.4▲), Iran (4.5), Iraq (8)
Non-reformed (low) Institution	“Engaged Non-reformers” Congo Dem. Rep. (6.3▲), Myanmar (17.8▲), Eswatini (2.7), Tajikistan (3.2), Venezuela (4.5)	“Unengaged Non-reformers” Cuba (3.4), Eritrea (6.9), Libya (2.8◇), North Korea (4), South Sudan (NA), Sudan (9.8), Syria (0.4), Turkmenistan (10.9), Uzbekistan (4.1), Yemen (4.6)

▲: Average GDP per capita under \$200 in 1991-1993, ◇: above \$5,000; Numbers in parentheses are fold-increase in exports from 1992 to 2015.

Table 2: Typology of Autocracies (Post-1990). *Note:* autocracies are roughly defined as those with average Polity ≤ 0 in 2000-20. Low institution-level refers to institutional levels that are below the thresholds for VDem PR protection and rule of law (see Appendix B.3). Together, “engaged reformers” accounted for over 90% of autocracies’ GDP in 2015. Countries like Equatorial Guinea grew from a very low base.

Empirical Implications

The analysis yields three testable predictions. First, reformed autocracies should gain most from post-1990 WTO accession relative to their pre-engagement baselines, exhibiting the largest accession effect on exports. Second, autocratic engaged reformers should outperform

²⁸ Author’s calculation of WDI data.

comparable democracies in absolute export levels. Additionally, as theorized in Section 4, autocratic advantage should be greatest at moderate levels of institutions and decline toward the low and high ends.²⁹ The logic stands on the post-1990 developmental paradigm shift, autocracy’s inherent disadvantages in trade-insulated economies, and WTO-centered trade integration. Then, these predictions, if true, strongly support the argument that globalization facilitates autocratic rise.

5 Empirics: Testing “Engaged Reformers”

The empirical tests focus on trade as the arena of demand competition and do not adjudicate among possible mediating mechanisms. It is nonetheless informative to first examine whether autocratic engaged reformers exhibit broad economic patterns the theory predicts. Table 3 shows that among developing WTO members during 2000-20, the group of autocratic engaged reformers is more industrial, attracts more greenfield foreign investment, and saves and invests at higher rates while consuming less.

Regime Category	Growth (%)	Indus. Output (%)	Greenfield FDI (%)	Fixed Inv. (%)	Saving (%)	Export (%)	Consump. (%)
Aut. Engaged Reformer	4.6	35.8	6.0	25.6	25.3	32.8	58.4
Dem. Engaged Reformer	4	25.4	5.3	22.2	15.3	25.3	69.7

Table 3: Descriptive Performance of Engaged Reformers (2000-20). *Note:* Averaged over in-WTO country-years among 133 developing countries (< \$20,000 average income in 2000). Institutional cutoffs are described below. Source: World Bank & fDi Markets.

5.1 WTO Accession

I next examine the effect of WTO accession—the most significant embodiment of trade integration. I empirically choose the cutoff year 1990 because it not only witnessed a dramatic global political shift—the end of the Cold War—but also an economic shift: the unprecedented proliferation of trade agreements and flows of goods and capital (Pandya 2016). Additionally, there had been worldwide domestic reforms and democratization. In Appendix C.10,

²⁹I show the results are robust to low/mid/high classifications (Appendix).

I calculate time-rolling estimates showing that the results are insensitive to specific cutoff years.

New WTO Joiners

Between 1990 and 2020, a total of 64 countries joined the WTO/GATT, and almost all were developing countries in 1990 (except Liechtenstein). Of them, 25 (Freedom House Index ≥ 8) or 18 (Polity ≤ 0) were autocratic states in 1991.³⁰ These countries do not account for the majority of existing autocracies, but include major autocracies such as China, Russia, Saudi Arabia, Vietnam, the United Arab Emirates, Qatar, Oman, Kazakhstan, Tajikistan, Kyrgyzstan, Bahrain, Tunisia, Angola, Laos, Cambodia, Venezuela, and Jordan. They account for over 90% of autocracies' total GDP and population. The spill-over effect is non-negligible: they significantly trade with non-WTO autocracies (Applebaum 2024). For example, Russia, China, or Saudi Arabia can more freely trade with Iran, Iraq, Cuba, and North Korea, while China's post-WTO rapid growth greatly contributed to the commodity boom during the 2000/10s, which benefited non-WTO autocracies (Hamilton 2009; Kilian and Hicks 2012).

Additionally, for semi-engagement, some non-WTO autocracies have been granted MFN status by the United States or the EU, such as Azerbaijan, Belarus, Serbia, and Turkmenistan. Others enjoy varied regional trade deals. Almost all remaining countries were granted WTO observer status.³¹ However, countries' export potential should remain largely constrained prior to joining the leading trade regime.

The Baseline: Gravity Model

I start with the gravity model, regarded as “one of the most robust empirical findings (Chaney 2018),” while absorbing recent methodological improvements (Eaton and Kortum 2012; Goldstein et al. 2007; Head and Mayer 2013; Rose 2004). Compared to the matching method below, the advantage is that the gravity model accommodates *continuous* Polity scores and provides a benchmark comparable to existing findings.

I first exploit variation across bilateral dyad-years to estimate the moderating effect regime

³⁰Russia's Polity = 3 in 1995.

³¹Observers must start negotiations within five years, implying they are trying to meet conditionalities and enjoying multiple benefits, such as speaking and learning opportunities, as well as signals to investors.

type on trade flows, specifically, the interaction $WTO \times Polity$.³² WTO membership should increase bilateral exports in similar directions, which sum up to aggregate patterns.³³ I specify various gravity models, including: 1) conventionally used WTO dummies *Both_WTO* and *One_WTO* in relation to the reference *Neither_WTO* (e.g., Goldstein et al. 2007; Rose 2004), 2) *Both_WTO* and *One_WTO* (exporter), and 3) exporter WTO dummy. I also add a model for imports. I control for a standard set of dyad-level covariates (see Table C.4) and an augmented set of importer-year, directed-dyad, and year fixed-effects, accounting for “multilateral resistance” effects (Anderson and van Wincoop 2003). The dependent variable is $\log(\text{exports} + 1)$. Trade data is drawn from CEPII’s Gravity dataset which aggregates data sources such as IMF DOTS and UN Comtrade with over 1,000,000 dyads across 70 years. Given the data and treatment structure, I conservatively cluster robust standard errors by exporter-year and dyad to account for intra-country and intra-dyad error-term correlations.³⁴ Institutions such as PR protection are not included to avoid possible post-treatment bias from WTO accession.³⁵

³²Since post-1990 regime types vary significantly less for most countries (see Appendix), I use Polity in 2000. Alternative measures such as Polity average (2000-20), real-time Polity, Freedom House index show consistent results. For pre-1990 years, only real-time Polity is used.

³³E.g., by influencing contract enforcement, institutional reliability, technology and management knowhow transfer, and trade costs, as well as importers’ trade barriers and trust in importing from exporter.

³⁴I avoid over-clustering by exporter, as error patterns for the same country in the 1990s may be much different from those in the 2010s (Yu 2010). Results are robust to exporter clustering.

³⁵Nonetheless, results hold with inclusion.

	1. Exports (FE)		2. Exports (FE)		3. Exports (FE)		4. Imports (FE)		5. Exports (CRE)	
	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990
<i>One WTO</i>	0.304*** (0.052)	-0.129* (0.061)								
<i>One WTO</i> $\times$ <i>Polity_i</i>	-0.003 (0.005)	-0.025** (0.008)								
<i>Both WTO</i>	0.615*** (0.084)	0.142 (0.092)	0.314*** (0.046)	0.271*** (0.047)					0.309*** (0.015)	0.208*** (0.012)
<i>Both WTO</i> $\times$ <i>Polity_i</i>	0.007 (0.006)	-0.041*** (0.010)	0.011** (0.004)	-0.033*** (0.006)					0.012*** (0.001)	-0.038*** (0.002)
<i>WTO_i Only</i>			0.306*** (0.052)	-0.036 (0.065)					0.304*** (0.017)	0.099*** (0.017)
<i>WTO_i Only</i> $\times$ <i>Polity_i</i>			0.005 (0.005)	-0.037*** (0.007)					0.004* (0.002)	-0.038*** (0.002)
<i>WTO_i</i>					0.307*** (0.042)	0.208*** (0.045)	0.050 (0.038)	0.053 (0.047)		
<i>WTO_i</i> $\times$ <i>Polity_i</i>					0.009* (0.004)	-0.033*** (0.006)	0.000 (0.003)	-0.011 (0.006)		
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Exporter Means									✓	✓
Dyad Means									✓	✓
Importer-year FE	✓	✓	✓	✓	✓	✓	✓	✓		
Dyad FE	✓	✓	✓	✓	✓	✓	✓	✓	RE	RE
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num.Obs.	220 706	540 324	220 706	540 324	220 706	540 324	215 680	530 191	216 643	523 818
R2 Adj.	0.868	0.886	0.868	0.886	0.868	0.886	0.866	0.879	0.873	0.882
R2 Within Adj.	0.035	0.012	0.035	0.012	0.035	0.012	0.042	0.021		
BIC	866 976.4	2 262 286.9	866 969.8	2 262 074.3	866 966.1	2 262 265.9	855 684.9	2 256 659.6	713 737.5	1 907 045.7

* p < 0.05, ** p < 0.01, *** p < 0.001

Table 4: The Effects of Joining the GATT/WTO. *Note:* Full specifications in Table C.4. Robust standard errors are clustered by exporter-year and dyad for FE models.

Table 4 displays the results of various operationalizations of GATT/WTO membership. By looking at the $WTO \times Polity$ interaction terms (in bold), the effect of the WTO on exports is larger for democracies pre-1990, but larger for autocracies post-1990.³⁶ For magnitude, WTO membership on average inflates exports by 13.9% for average autocracy ($Polity = -5$), while decreasing exports by 16.4% for average democracy ($Polity = 5$). For comparison, autocracies do not import more (group 4), consistent with the export-specific nature of autocratic advantages. One explanation for the negative effect for democracies may be that trade opening-up is subject to more import competition. The results are robust to importer type (whether being prior-1990 joiners, regime type). Figure C.10 visualizes the rolling estimates for WTO membership from 1948 to 2020.

I conduct sensitivity tests following Cinelli and Hazlett (2020) gauging how strong an omitted confounder needs to be to completely explain away the effect of, i.e., $WTO \times Polity$. Figure C.11 plots the sensitivity contours which represent the estimates of $WTO \times Polity$ given the hypothetical partial R^2 of the omitted confounder with treatment ($R^2_{D \sim Z|X}$) and

³⁶The post-1990 interaction effect is moderately smaller if removing China-, or Russia-, or thirteen OPEC-origin dyads, but becomes close to zero if all are removed, which is nonetheless better than the pre-1990 negative effect. Note that removing all that account for the majority of autocracies' GDP (over 90% in 2015) raises representation bias. More importantly, removing all doesn't affect subsequent tests of "engaged reformer."

outcome ($R_{Y \sim Z|D, \mathbf{X}}^2$). Any omitted confounder that nullifies the estimates would need to be over 250 times and 1000 times as strong as GDP and population with both treatment and outcome, respectively.³⁷ Additionally, I estimate a Correlated Random Effects (CRE) model, including covariate group means to allow correlation between random effects and regressors, recovering coefficients on time-invariant variables otherwise absorbed by dyad fixed effects. Following Santos Silva and Tenreyro (2006), I re-estimate the gravity model using PPML to address potential bias from heteroskedasticity by OLS. As shown in Column 9&10 of Table 4, the result is similar to the fixed-effects model.

Autocratic Engaged Reformers

As argued in Section 4, the regime-moderated WTO effect varies across institutions, and autocratic advantages from discretionary institutions should amplify exports the most at moderate levels (i.e., U-shaped). I operationalize domestic institutions by the rule of law and PR protection, respectively (both from VDem). Institutional levels are divided into three categories: low, moderate, and high.³⁸ Since the actual post-accession reform trajectory matters, I measure institutional levels by “10-year average institutional levels” after accession and assign dyads into the corresponding institutional categories.³⁹ I do not control for institution in the model for possible post-treatment bias: institutional change may be partly affected by WTO membership.⁴⁰ I combine low and high coded as “other” and code moderate range as “reformed.”⁴¹ However, results are robust to binary categories (i.e., low/moderate-high) and three categories (i.e., low/moderate/high, see Appendix). I fit a model including a three-way interaction among post-1990 joiners:

$$Export_{ijt} = \beta WTO_{it} \times Polity_i \times Institution_i + \delta \mathbf{X}_{ijt} + \gamma_{ij} + \eta_t + \epsilon_{ijt}$$

³⁷As noted by Cinelli and Hazlett, these results are conservative for the case of multiple (possibly non-linear) omitted confounders.

³⁸I calculate thresholds combining lowest/highest 20 percentiles of institutions among less developed countries (GDPPC < \$20,000 in 2000, see Appendix for details). The final ranges are (0.2, 0.7) for rule of law and (0.3, 0.8) for PR protection.

³⁹This measure captures the theory-required reforms alongside accession, but may introduce post-treatment bias. Results hold using pre-WTO institutional levels.

⁴⁰However, results hold when controlling for institution.

⁴¹Autocracies (Polity < 0) are empirically rare above the moderate range (0.7 in the rule-of-law case), establishing the upper bound of common support.

where $Institution_i$ is a categorical variable of post-WTO 10-year averages of institutional levels (low, mid, high) of country i , and $\mathbf{X}_{ijt}$ denotes dyad-level covariates. γ_{ij} and η_t are dyad and year fixed-effects.

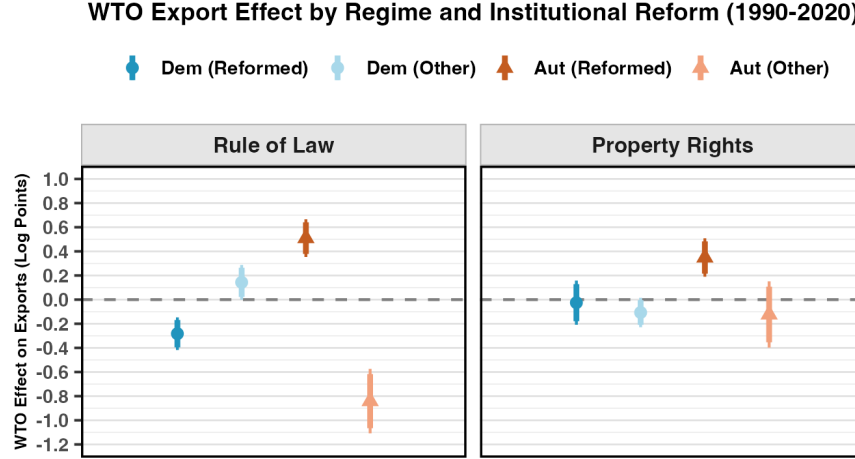


Figure 6: The Effects of Joining the WTO Conditional on Institutional Levels. *Note:* The y-axis means the WTO effects in percentage points.

Figure 6 shows that, as expected, autocratic reformers outperform all other regime categories. For example, *WTO* accession increases their average exports by 0.51 (nearly 67%) and 0.35 (42%) for rule of law and PR protection, respectively. Robustness tests include (1) varying institutional thresholds up and down 5 percentage points across a range of ± 10 (Figure 7), (2) changing post-WTO institution averages to 5 or 15 years, (3) binning by pre-WTO institutions, (4) leave-one-country-out, (5) using the World Bank rule-of-law index, and (6) using a dichotomous Polity (see Appendix). Despite the large dyad sample, given limited post-1990 WTO joiners, I use B-spline interactions and a generalized additive model (GAM) to flexibly estimate the expected U-shaped regime effect across institutions (Appendix C.14), with a wild cluster bootstrap to verify standard errors. I also run models with the variable *both-WTO* (i.e., within-exporter treatment variation) and a CRE model, all of which yield similar results (see full table in Table C.5).

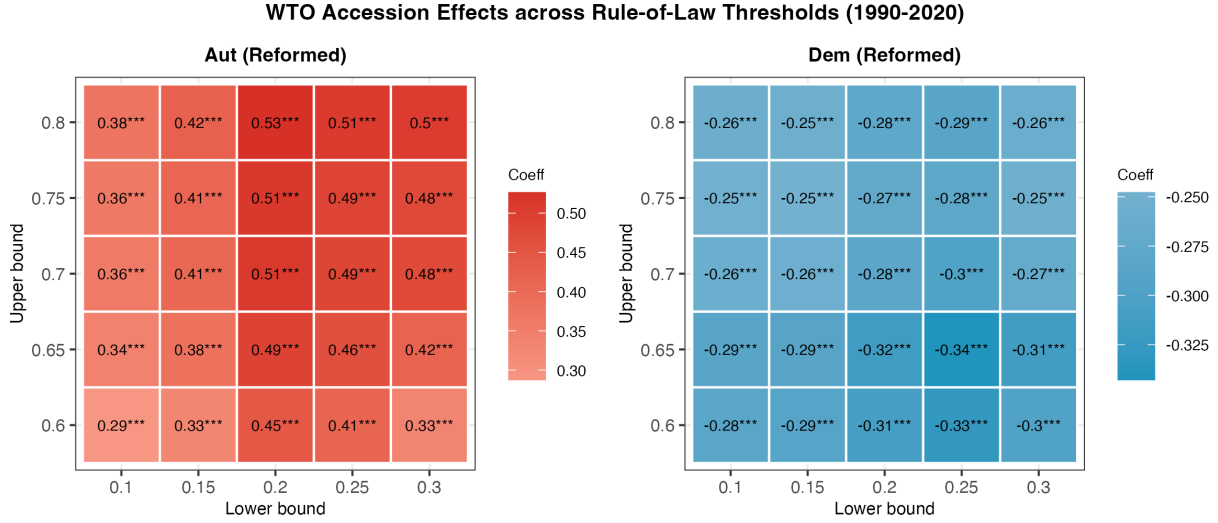


Figure 7: WTO Accession Effects across Rule-of-Law Thresholds (1990–2020). *Note:* Each cell reports the estimates in exports from WTO accession under alternative rule-of-law classification cutoffs. The lower (upper) bound defines the maximum (minimum) institutional levels. Baseline: lower = 0.2, upper = 0.7. *** $p < 0.01$.

The most significant challenge is the selection of autocratic engaged reformers who export more due to other attributes. The empirical models address this directly by controlling for numerous pre-treatment covariates such as income levels and institutional quality. Moreover, the central comparison is between autocratic and democratic engaged reformers both of which selected into the same structural configuration—reform and integration—as the minimal threshold that does not guarantee export competitiveness. The residual gap is, I argue, what regime characteristics predict (see Section 6 for mechanism evidence).

The reasons of no pre-1990 autocratic advantages, I argue, are that there had been neither substantive institutional reforms worldwide (or only on paper for some newly decolonized states) nor globalization. In other words, the scope condition had not been sufficiently realized. While some autocracies and democracies did trade before 1990, the scale, depth, and institutional environment were categorically different from the globalized economy. The global value chain had not taken off—for example, South Korea largely relied on developing indigenous supply chains (Baldwin 2016).

Staggered Difference-in-Differences (DiD) with Matching

In addition to model-based approaches, I use staggered DiD with matching as a nonpara-

metric identification strategy to estimate the effect of WTO accession. Apart from directly testing country rather than dyad levels, it offers significant advantages over, e.g., fixed-effects methods for panel data (Imai et al. 2022), by explicitly constructing counterfactuals through matching on pre-treatment covariates.⁴² This approach ensures comparable comparisons and reduces outlier influence, providing more robust causal estimates. It also provides insight into the long-term effects, as the WTO effect may persist over time. The ATT (Average Treatment effect on the Treated) estimator is expressed as follows:

$$\hat{\tau}^{ATT} = \frac{1}{\sum_{i=1}^N \sum_{t=L+1}^{T-F} D_{it}} \sum_{i=1}^N \sum_{t=L+1}^{T-F} D_{it} \left\{ (Y_{i,t+F} - Y_{i,t-1}) - \sum_{i' \in \mathcal{M}_{it}} w_{it}^{i'} (Y_{i',t+F} - Y_{i',t-1}) \right\}$$

where D_{it} is treatment indicator (1 at treatment onset). $Y_{i,t}$ is the outcome for treated unit i at time t . $\mathcal{M}_{it}$ is the set of matched control units for treated unit i at time t . $w_{it}^{i'}$ is the weight for control unit i' matched to treated unit i .

Specifically, Covariate Balancing Propensity Score (CBPS) weighting is used to balance covariates.⁴³ CBPS estimates propensity scores such that covariates are balanced (Imai and Ratkovic 2015). Weighting methods are particularly effective in non-large datasets because they retain all available control units. As DiD is inconvenient for handling interaction effects, units are roughly stratified into democracies ($\text{Polity} \geq 0$) and autocracies ($\text{Polity} \leq 0$).⁴⁴ For all tests, I use export (log) as the DV, similar to gravity models.

Utilizing the country-year panel dataset, I set lags $L = 4$ to match pre-treatment histories and $F = 5$ for forward effects, as joining the WTO may not immediately boost trade.⁴⁵ I focus on pre-treatment covariates that theoretically affect both WTO accession and exports,

⁴²PanelMatch is better suited to this setting than event-study in Callaway and Sant’Anna (2021) due to thin treatment cohorts and likely pre-treatment imbalances.

⁴³I choose among mahalanobis matching, propensity score matching/weighting, and CBPS matching/weighting for the best performance in balancing covariates. The standardized mean differences (SMD) of most covariates are within the rule-of-thumb of 0.2 (see Figure C.12).

⁴⁴As regime types have remained relatively stable since the mid-1990s, I capture regime type in 2000 for the same country throughout the period.

⁴⁵Longer leads and lags are refrained from, as they can eliminate more units that don’t match.

including GDP (log), GDP per capita (log), Polity, population (log), race (white), geopolitics (NATO membership), natural resource intensity (% of GDP), industrial output intensity (% of GDP), rule of law, and lagged exports (for pre-WTO export-level self-selection). I avoid controlling for direct post-treatment covariates such as tariff rate, which is lowered upon WTO accession.⁴⁶

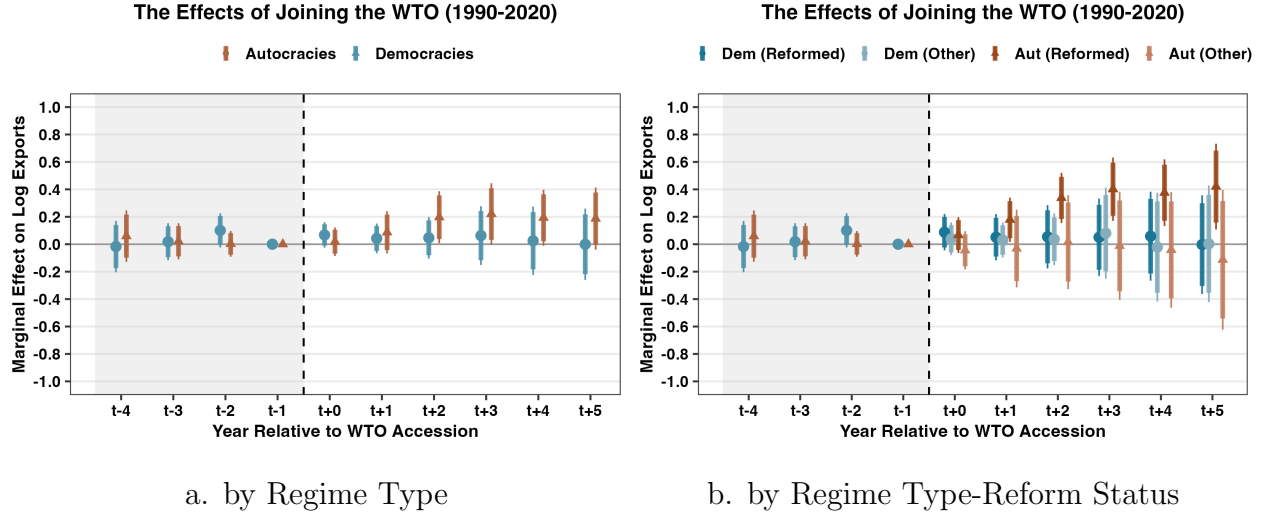


Figure 8: Effects of Joining the WTO on Exports (Post-1990). *Note:* Autocracy: Polity ≤ 0 . The WTO effect of autocracy (and engaged reformer) is statistically significant while those of others are not. Placebo periods show that parallel trends assumption hold (the shaded area, with t-1 as reference time). Standard errors are estimated with bootstrap.

Figure 8 plots the WTO effects for democracy and autocracy, respectively. Panel a shows that from t+3 to t+5, autocracy's WTO effect is approximately 0.2, while democracy's is insignificant. This is highly consistent with gravity models (Table 4), which report that the effect difference between average autocracy (Polity = -5) and average democracy (Polity = 5) is approximately 0.3 for the post-1990 period. Panel b reports that autocracy's effect is primarily driven by autocratic engaged reformers.⁴⁷ Results are robust to leave-one-country-out checks (Appendix).

In Figure 9, I show a few representative countries—China (the largest autocracy), Angola (a resource-rich African autocracy), Ukraine (a post-Soviet democracy), and Costa Rica (a

⁴⁶I match rule of law because WTO conditionality (post-treated) is more about trade-related liberalization and intellectual PR protection (Allee and Scalera 2012). Results hold for no matching.

⁴⁷Same as above, rule of law between 0.2 and 0.7.

development success in Latin America)—using the synthetic control method (SCM). China, for example, compared to the control unit, had 189% and 242% more exports 10 years and 15 years after WTO accession, respectively.

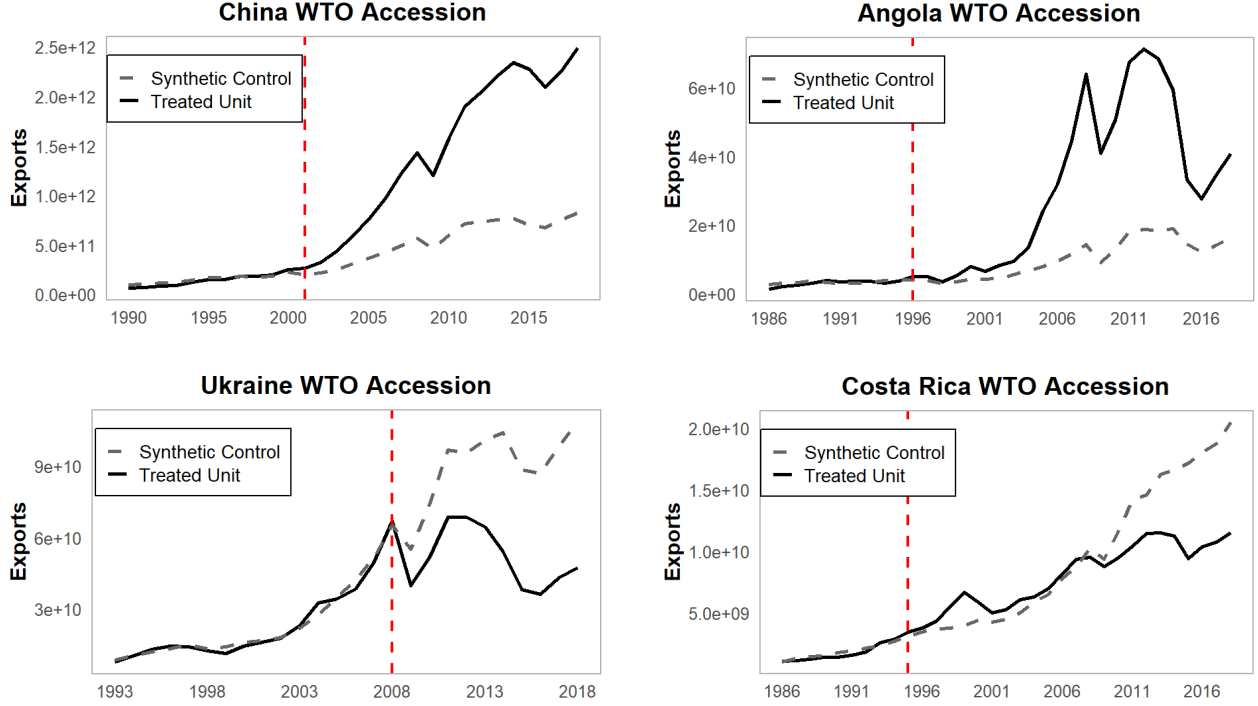


Figure 9: Examples by Synthetic Control Method. *Note:* Red dashed lines denote WTO-joining years. Covariates to predict the control unit include: GDP, GDP per capita, GDP growth, population, trade openness, Polity, PR protection, resource intensity, industrial intensity, and lagged exports.

This contrast suggests that autocracies likely have much weaker export performance absent trade engagement, because autocratic institutions usually suppress domestic demand making external demand competition essential to their growth model. Trade isolation forecloses precisely this opportunity. In Appendix F.1, I show that domestic reform alone yields much weaker effects during their non-WTO periods.

Additionally, I construct an engaged-reformer status as a time-varying indicator, equal to one when a country is simultaneously a WTO member and institutionally within the moderate range. I estimate its effect using the same matching-based event-study design. Appendix D details the construction and estimation, which confirm the pattern above.

5.2 Absolute Levels of Trade Performance

To test whether the regime category of autocratic engaged reformer leads to higher absolute export levels under a globalized economy, I estimate full gravity models on the pooled dyadic sample (169 countries, 532,574 dyad-years over 1990-2020) and divide all dyads into four regime categories (WTO $\times$ Reformed/Other) based on exporter.⁴⁸ Models are estimated separately within each cell by weighted least squares with inverse-density weights.

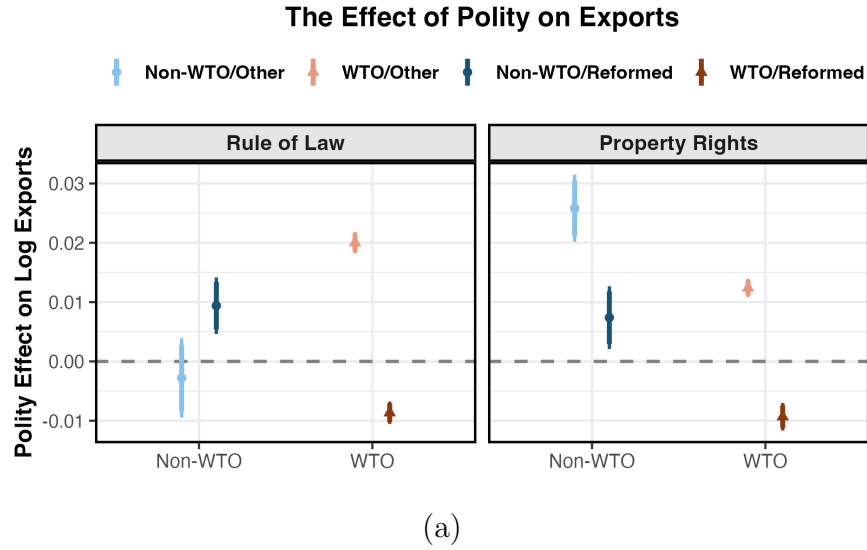


Figure 10: Polity Effect on Exports by WTO Status and Institutional Quality. *Note:* Coefficients from gravity regressions of log bilateral exports on polity score, estimated separately for WTO/Non-WTO countries and institutional reform status.

Figure 10 reveals a consistent pattern across both institutional measures. Among WTO members with moderate institutional levels (WTO/Reformed), being more autocratic is associated with significantly higher exports—the polity coefficient is negative. This autocratic advantage does not extend to WTO members outside the moderate range (WTO/Other) where the polity effect turns positive, nor to non-WTO reformed countries (Non-WTO/Reformed) where democracy favors exports. The results thus confirm the theoretical expectation: the export premium of autocracy is specific to countries that combine WTO membership with moderate institutional reform, which are the defining characteristics of autocratic engaged

⁴⁸I use the same institutional categories as the above tests for rule of law and PR protection (lagged).

reformers.

6 Mechanisms

The previous evidence established aggregate export patterns consistent with the regime mechanisms, for which this section further triangulates across two complementary layers. I first present a sectoral analysis that verifies the theory’s sector-specific expectations while ruling out alternatives. I then go deep into the mechanisms at work: evidence from Japanese investors in Vietnam and the Philippines that validate firm-level micro-foundations, as well as a paired comparison of China and India that traces the mechanisms to specific government policymaking.

6.1 Sectoral Patterns: beyond a single story

Sectoral tests examine whether export gains concentrate in sectors where regime characteristics, such as centralized coordination and lack of constraints, should matter most, while ruling against a simple narrative of engagement, catching-up, “race-to-the-bottom” in labor-rights, or resource only. For example, coordinated mercantilist policies or wage suppression may favor manufactured and intermediate exports that reflect GVC participation more than agricultural and primary goods. To illustrate, China shifted millions of laborers from agricultural sectors to manufacturing after its WTO accession (Erten and Leight 2019).

The UNCTAD classifies sectors based on manufacturing factors (labor, natural resource, and technology).⁴⁹ Accordingly, I create four categories of product types: agriculture, natural resource, basic manufacturing, and mid-to-high manufacturing based on manufacture types at the broad Harmonized System (HS2) level. The World Bank’s WITS on the other hand, classifies Harmonized System sectors into raw materials, intermediate goods, consumer goods, and capital goods based on GVC participation. I merge six-digit HS code into the four broad categories. I then ran sector-level gravity models with the interaction term ($WTO \times$

⁴⁹See https://unctadstat.unctad.org/EN/Classifications/DimSittcRev3Products_Tdr_Hierarchy.pdf.

Reformed/Other) using CEPII's BACI data.

Figure 11 presents consistent results across both product classifications (converted to percentages). Reformed autocracies show large positive effects in every sector, from roughly 125% in agriculture to over 275% in intermediate goods. The advantage is absent for reformed democracies, confirming that performance is specific to the intersection of regime type and domestic reform.

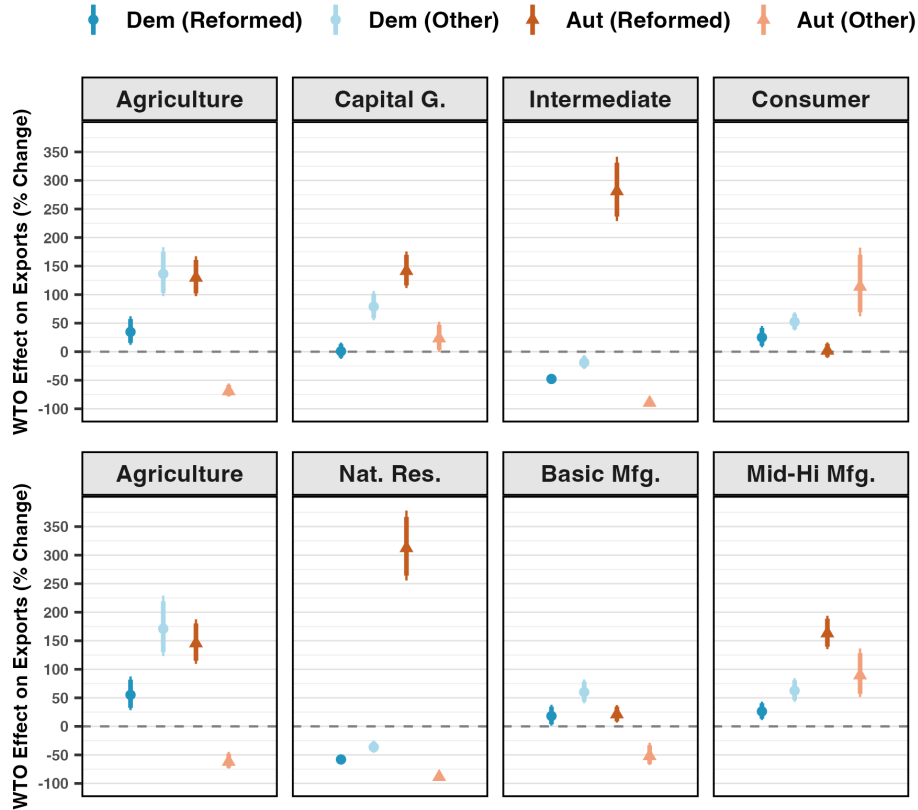


Figure 11: WTO Effect across Sectoral Categories (1990-2020). *Note:* Estimated WTO accession effects on bilateral exports (% change) across four regime-reform groups and two HS2 sector classifications. Gravity models with dyad and year fixed-effects; standard errors clustered at the dyad level.

The sector-level pattern offers additional leverage. Autocratic advantages are comparatively small in agricultural, consumer, and basic manufacturing goods, where state coordination is least decisive. The largest effects appear in capital goods, intermediates, and mid-to-high manufacturing apart from natural resources, consistent with FDI-driven upgrading, integration into the GVC, and coordinated industrial policy (echoing the descriptive compar-

isions in Table 3), rather than labor cost competition alone. The effect, moreover, extends well beyond natural resources, effectively ruling out a commodity-driven explanation. Results are robust when Polity is treated as continuous: the WTO export effect declines monotonically as regime type becomes more democratic across all sector types (Appendix).

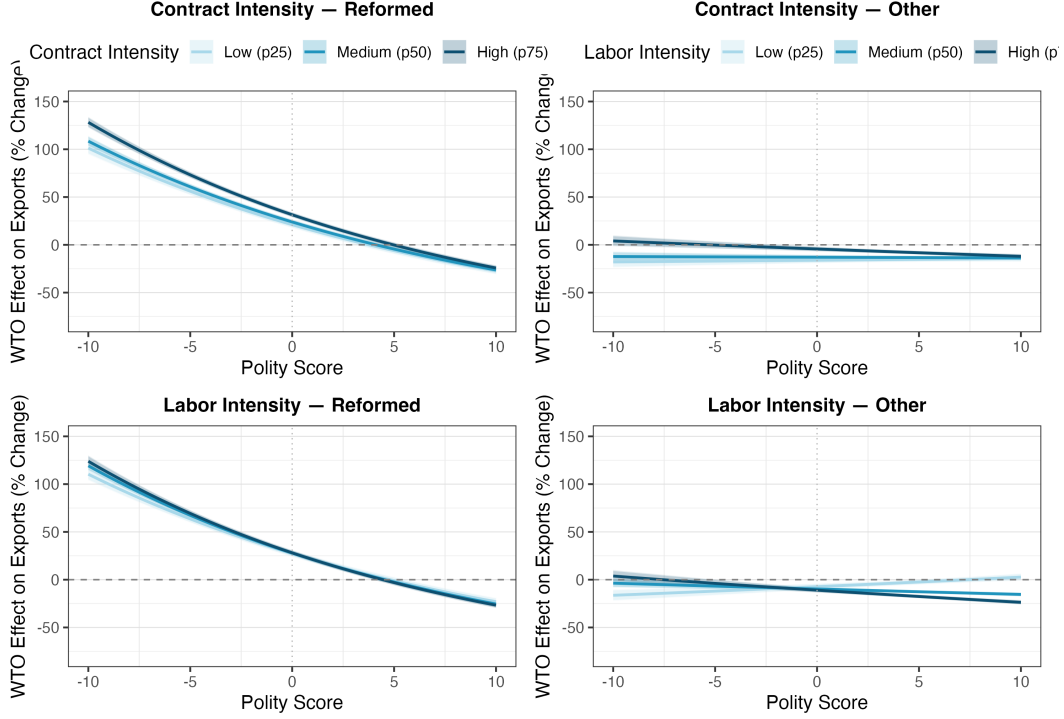


Figure 12: Contract and Labor Intensity Heterogeneity in WTO Effects (1990-2020). *Note:* WTO accession effect on bilateral exports across Polity scores at the 25th, 50th, and 75th percentiles of contract intensity (Nunn 2007) and labor intensity (NBER-CES), measured at HS2 level. Gravity models with dyad-HS2 and year fixed effects; standard errors clustered at dyad-HS2.

Figure 12 plots the WTO effect across Polity scores at low, medium, and high levels of contractibility (Nunn 2007) and labor intensity (NBER-CES) for reformed and unreformed countries separately. For reformed joiners, contract intensity favors autocracies and higher level generates a positive premium, consistent with state coordination in relationship-specific sectors (Nunn 2007). The effects by labor intensity degrees are larger for autocracies, but shows little differentiation—suggesting their broad wage advantage. Unreformed autocracies show much smaller overall effects with modest labor-intensity differentiation, suggesting their gains are more narrowly labor-cost driven. Their contract-intensity premium may reflect co-

receive enforcement capacity or simply their resource-heavy sectors that score high on contract intensity.

6.2 Investors and Governments: Mechanisms at Work

Japanese Investors in Vietnam and the Philippines

Private firms (the main actors in the post-1990 GVC era) are what the theory ultimately runs through: regime features matter only because firms perceive them as credibility or cost advantage, and relocate production accordingly. I examine Japanese investors' preferences regarding Vietnam and the Philippines—a comparable pair—for those making real investment decisions. In the 1980s, both countries shared similar income levels (both below \$600), economic structures, demographic profiles, geographic conditions, ethnic compositions, and resource endowments. Their export trajectories resembled each other until the early 2000s (Figure G.20).

Vietnam initiated economic reforms in 1986, and by 2001 had obtained MFN status from most Western countries. Although running persistent trade deficits before 2010, Vietnam had implemented a state-led, mercantilist development strategy modeled in part on China's. The country's ability to pursue consistent policy agendas, long-term planning, and gradual and managed liberalization has been widely attributed to its centralized political system (Kirkpatrick et al. 2001). Vietnam's WTO accession in 2007 marked a turning point, and within five years, it ran sustained trade surpluses. In contrast, the more democratic Philippines witnessed elites dominating the democratic process and capturing rents, while resources were diverted away from investment in human development and infrastructure, key to export performance (Baulch 2016).

Table 5 collects and summarizes six rounds of data based on JETRO's annual "Survey of Japanese-Affiliated Firms in Asia and Oceania" to provide firm-level evidence that investor perceptions translate into actual investment behavior.⁵⁰ The aggregate export patterns in

⁵⁰JETRO (Japan External Trade Organization) is a government-affiliated agency that annually surveys Japanese-affiliated firms.

Table 5: Japanese Affiliate Survey Evidence: Vietnam vs. Philippines, 2009–2019

Metric	Country	2009	2011	2013	2015	2017	2019
<i>Panel A: Japanese FDI Presence (firms in JETRO database)</i>							
Firms in JETRO database	Vietnam	277	292	585	1,027	1,345	1,608
	Philippines	333	230	237	239	382	517
<i>Panel B: Survey Responses (% of responding firms; N in parentheses)</i>							
Expansion intent (%)	Vietnam	77.6 (76)	68.0 (150)	~65 (435)	63.9 (557)	69.5 (652)	63.9 (855)
	Philippines	<50 (115)	43.0 (121)	~55 (150)	55.1 (119)	63.4 (73)	51.8 (139)
Wage hike: top challenge (%) [rank among problems]	Vietnam	85.2 [#1]	83.3 [#2]	78.2 [#1]	77.9 [#1]	75.2 [#1]	72.0 [#5]
	Philippines	<top 3	48.4	41.8 [#5]	31.4	45.8 [#5]	44.5
Production (goods) as expansion function (%)	Vietnam	—	—	38.3 [#2]	42.4 [#1]	35.9 [#2]	—
	Philippines	—	—	30.6	23.1	16.3	—
BPO/Admin services as expansion function (%)	Vietnam	—	—	low	low	low	—
	Philippines	—	—	high	44.4 [#1]	44.4 [#1]	—

Source: JETRO Survey of Japanese-Affiliated Firms in Asia and Oceania, 2009–2019. Panel A counts all Japanese-affiliated firms ($\geq 10\%$ Japanese equity) registered in JETRO’s database; Panel B percentages are computed over responding firms only. $\sim$ = approximate; — = not disaggregated.

Section 4 leave open whether firms consciously respond to the governance mechanisms theorized here. The JETRO data provides partial answers. Panel A shows Japanese firm presence in Vietnam growing nearly sixfold between 2009 and 2019, while Philippine presence barely grows. Panel B reveals two patterns consistent with the mechanisms: Vietnam’s firms report persistently higher expansion intent, and wage hikes rank as the top operational challenge—suggesting that as foreign firms pour in, a tighter labor market began threatening its competitive labor costs in the first place (partly due to weak labor-rights). Vietnamese expansion is concentrated in goods production rather than the services and BPO functions that dominate Philippine operations, consistent with Vietnam’s mercantilist, manufacturing-oriented investment climate.

To test whether the four mechanisms in Section 3 manifest in investor discourse, I conduct a systematic content analysis of Japanese financial news coverage of investing in Vietnam and the Philippines. I collect keyword-filtered articles from the Nikkei, Daily Yomiuri, and related outlets via Factiva spanning 2001–2020, yielding over 10,000 articles coded as relevant to investment in either country. Each article is scored by GPT-4o-mini on five dimensions corresponding to four theoretical mechanisms—centralized power, weak institutional constraints, limited accountability, mercantilist mentality, plus overall attractiveness—using both a continuous salience scale (1–5) and a binary presence indicator (0/1). To validate

LLM coding reliability, I independently code a random subsample of 100 articles, yielding a high inter-rater agreement. I then estimate two OLS models regressing each dimension score on a Vietnam indicator, with year and outlet fixed-effects, and log article length as controls with robust standard errors.

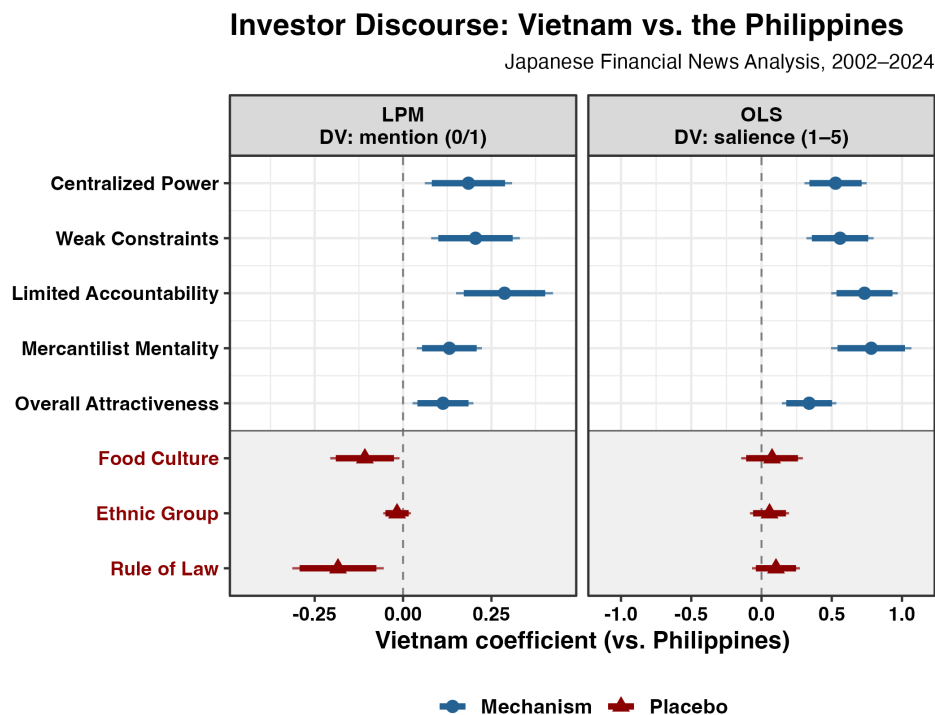


Figure 13: Investor Perception Analysis. *Note:* data retrieved from the articles of major Japanese Financial Newspapers. Scores are rated by LLM (GPT-4o) with human validation.

In Figure 13, Vietnamese articles score significantly higher than Philippine articles on all five dimensions across both the continuous and binary specifications. For example, the limited accountability mechanism—proxied by labor cost competitiveness and suppressed labor rights—where Vietnam articles score 0.73 points higher on the salience scale (panel b) and are 28.7 percentage points more likely to contain this framing (panel a). To assess whether the results reflect investment-relevant perceptions rather than general country-specific bias, I add three placebo dimensions: food culture, ethnic group, and rule of law. The result shows no systematic Vietnam advantage for placebos.

Comparative Case: China and India

The sectoral pattern is consistent with the mechanisms but does not observe them in operation, which requires within-case causal-process evidence, preferably in an otherwise similar engaged-reformer pair. The China-India dyad between the largest autocracy and democracy is not only significant, but it indeed supplies a comparable pre-globalization baseline: before the 1980s, the two were not identical but shared many structural similarities—demographics, coastal geography, labor and resource endowments, and inward-looking economies (India’s extensive system of licenses, permits, and regulations, known as “License Raj,” parallels China’s planned economy). For decades, export paths tracked closely until China’s “engagement-and-reform” turn.

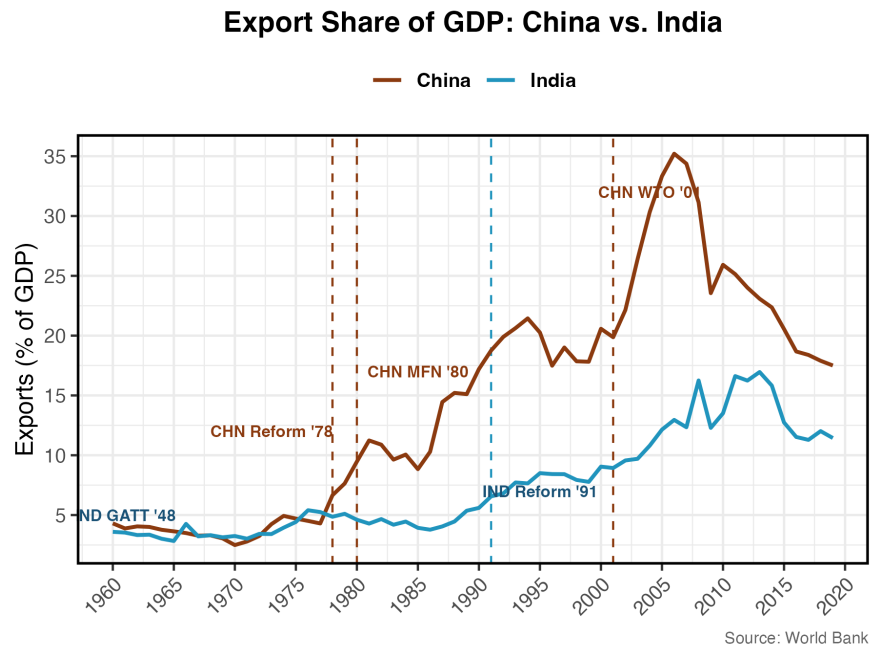


Figure 14: Historical Exports (%) Trajectories: China and India.

China fits the theory and its scope conditions precisely, and the mechanisms operate through decisions that bypass would-be veto points. Market-oriented reforms began in 1978; against bureaucratic and ideological opposition, the central leadership created the Special Economic Zones by fiat in 1979-80, carving out export enclaves with flexible labor and tax regimes without obstruction (Vogel 2011; Shirk 1993). Semi-engagement followed—MFN access to the U.S. market in 1980, alongside heavy World Bank borrowing—and exports took off even as trade deficits persisted (1980-1995). When reform stalled after 1989, Deng

Xiaoping’s 1992 “Southern Tour” restarted it through personal authority alone, an instrument a democratic executive hardly holds. The 1994 exchange-rate unification then produced a more competitive renminbi—a mercantilist tilt enacted with no independent central-bank, legislative, or interest-group constraint (Naughton 2007). WTO accession in 2001 delivered a further boost and sustained surpluses thereafter. Throughout, *centralized authority* enabled a gradualist, coordinated, and rapidly responsive approach (Lin et al. 2003), while *weak institutional constraints* permitted an openly mercantilist posture: managed exchange rates, export-processing zones, and directed credit (Huang 2008). *Limited accountability* sustained the suppression of labor-rights that underwrote cost competitiveness (Gallagher 2005).

By contrast, India’s attempts to copy these instruments were repeatedly constrained at democratic veto points: rigid labor laws survived reform efforts, while land acquisition for industrialization and SEZs generated farmer mobilization, litigation, and political backlash, as seen in Singur and Nandigram (Besley and Burgess 2004; Jenkins et al. 2014). Despite having a higher nominal PR protection than China (0.77 vs. 0.35 in 2000), India’s democratic system, though more inclusive, must navigate coalition politics, public dissent, and legal constraints, resulting in slow decision-making and limited discretionary practices (Groth 2006). As such, India had the same playbook; its institutions would not let governments run it. India’s GATT signatory status back in 1948 didn’t stimulate trade much. Its export performance diverged from China’s around 1980, with persistent trade deficits.

A competing explanation—that China’s export surge reflected Hong Kong, Taiwan, and diaspora FDI rather than regime characteristics—understates how such capital was institutionally channeled. SEZs, credible policy commitments, and labor flexibility helped direct external capital into export-oriented manufacturing. India’s large diaspora was never mobilized to the same extent, itself a function of the policy environment.

However, conditional on similar structural within-country factors, Modi’s authoritarian and centralizing turn varies the institutional configuration alone (Jaffrelot 2021); New Delhi then moved more towards state-directed, mercantilist policies, and electronics exports rose fivefold to \$22 billion from 2018 to 2023.⁵¹ Although the timing coincided with the “China+1”

⁵¹ “10 Years of Make in India,” Government of India, 2024.

relocation window, that opportunity was available to many democracies and to India throughout; India captured it only once centralization delivered the fast-tracked approvals, subsidies, and land that earlier governments, as Singur suggested, struggled to coordinate. The exogenous shock supplied the opportunity; the institutional configuration converted it into exports. Within the limits of a two-case design that cannot separate all mechanisms, the case shows where the mechanisms are operating—and failing to operate—as the theory expects.

7 Alternative Explanations

Catching up

Less developed countries (LDCs) may converge with developed economies through faster growth (Luo and Przeworski 2019). Yet, post-1990 WTO-joiners were mostly LDCs. Neither did autocracies start low. The correlation between GDP per capita (pre-WTO accession or in 1990) and Polity for post-1990 WTO-joiners is negative ($r = -0.25$ and -0.43)—autocracies started higher. Absolute trade levels also favor autocracies controlling for income per capita. Moreover, catch-up cannot predict why reformed and non-reformed autocracies move in opposite directions.

Importing Credibility

Arias et al. (2018) argue that autocracies benefit more from joining international institutions since they rely more on external credibility by importing institutions. However, this theory cannot explain differentiated performance between reformed and other autocratic WTO members. It is the combination of institutional reform, economic engagement, and autocratic institutional advantages that matters. Historically, domestic institutional improvement results much from active domestic reform pushed by governments.

Commodity Boom

What about the commodity boom and the spillover effect? I first delete the boom years (2004–2014) and OPEC countries, and the results hold. The spillover of joiners (e.g., China and others) and the buildup of a global commodity market do matter. However, this second-order

effect does not negate my argument that globalization facilitates autocratic rise. Plus, we do not observe autocratic advantages in the previous oil boom (1970s/80s), and resource-rich countries that do not meet the scope condition nonetheless underperformed (e.g., Venezuela, Iran, and Iraq). Gulf successes may be less remarkable absent structural changes in globalized commodity markets, and globalization-driven demand, investments, and technology diffusion. In fact, oil industries are capital- and technology-intensive.

Other Discussions

Trade integration and institutional reforms are arguably the two most important determinants of trade performance identified in the growth and development literature. Nonetheless, below I discuss some potential questions.

A story of China, Vietnam, Russia, and oil states? None of the descriptive data in Section 2 shows they are outliers. Moreover, the theory and empirical evidence of engaged reformer suggest that regime type plays an important role. Even if one insists that the theory fits better with these major autocracies, it largely explains autocratic rise.

State capacity. State capacity plays a pivotal role in economic development (Acemoglu et al. 2015; Dincecco 2017). Yet the association between high-performing states and strong capacity does not necessarily imply causation, that is, conditional probability $P(\text{capacity} \mid \text{performance})$ is not $P(\text{performance} \mid \text{capacity})$. However, state capacity alone is insufficient to explain why autocratic advantages emerged primarily after deep trade integration and market-oriented reforms. In the Appendix, I test the robustness of the results against broader measures of state capacity.

Semi-engagement. some autocracies are granted as part of PTAs, for example, U.S.-Vietnam or U.S.-Lao Bilateral Trade Agreements, which are controlled for in the models. Moreover, MFN is a core WTO principle and semi-engagement is part of globalization, while WTO membership provides much more benefits than a revocable MFN status. More importantly, any export gains from pre-accession MFN are absorbed into pre-treatment trends, meaning the estimated WTO effect is likely a lower bound on the full effect of institutional trade engagement.

Back-and-forth trade along the GVC may double count exports. However, democracies are usually more GVC-integrated, their gross exports may overstate true trade performance relative to autocracies. Moreover, external balance—net exports that net out intermediate inputs—also favors autocracies. Gross exports also remain the conventional benchmark for trade performance.

Lastly, although not the focus of the paper (demand competition), export-oriented FDI, rather than service-oriented, can boost exports (Helpman 1984; Pandya 2016). As export-oriented FDI usually follows globalized production (Helpman 1984; Markusen 1984), it is better understood as a post-treatment mediator or channel: it follows trade engagement rather than preceding it (Carnegie 2014). While the greenfield FDI data (fDi Markets) is only available since 2003, the pattern of higher inflows among autocratic engaged reformers (Table 3) and Japanese investors’ study are consistent with the mechanism.

8 Conclusion

The debate over regime type and economic performance has long hinged on the mechanisms through which political institutions shape development. This paper advances that debate by addressing a specific puzzle: why has the effect of regime type on trade performance changed so dramatically in the post-1990 period? Why can China and Vietnam succeed? I argue that economic globalization constitutes a necessary condition for autocratic trade advantages—one that prior scholarship has underweighted. Because autocratic regimes often lack inclusive institutions for growth, resulting in weak domestic demand (Acemoglu and Robinson 2012), autocracies are systematically more reliant on export-led growth. Leveraging WTO accession as a source of exogenous variation, the empirical findings confirm that autocratic engaged reformers systematically outperform democratic counterparts on export performance—a gap not explained by pre-treatment differences or selection into engagement.

The findings carry several theoretical implications. First, they qualify institutional theories of growth: while inclusive institutions matter for domestic demand-driven development, external demand can substitute for—and even reward—the exclusionary institutional ar-

rangements characteristic of autocracies. Second, they challenge optimistic accounts of trade liberalization. Liberal internationalists have argued that open trade fosters interdependence and democratic convergence; this paper shows that the existing trade regime has created conditions under which autocracies can thrive without political liberalization. Third, the findings speak to Downs et al.'s (2000) accession theory: sequential accession has proven insufficient to ensure compliance with WTO principles, and as autocratic members have thrived, the organization's capacity to function as a rule-based institution has eroded.

The trend depicted in the paper posts a sober future for the current liberal order and democracies. Some argue that democracies should establish their own trade bloc (Friedberg 2025). Whether framed as a contest between democracy and autocracy or as new great power rivalry, my findings indicate that global market forces – shaped by the existing trade system—may continue to favor autocracies. Given this dynamic, the continued deindustrialization of advanced democracies may not be surprising. The fact that economic globalization has produced this outcome is both unexpected and undesirable. Ultimately, where the global trade system and globalization should be headed hinges on its expectations and consequences; as Keohane (1984) posits, the means are justified by the ends. Future research should further explore the institutional advantages of reformed autocracies and whether democracies truly cannot compete on an equal footing.

Should countries embrace autocracy? The autocratic advantage identified here is conditional: it operates only through engagement and reform, and only under conditions of sufficient globalization—as the absence of pre-1990 effects confirms. Should deglobalization deepen through tariff escalation or regional bloc consolidation, the scope conditions enabling these advantages may attenuate. Moreover, institutional consistency may be hard to sustain across generations among autocracies. Future research should examine whether democracies face inherent institutional disadvantages in export competition under deep globalization, and whether the political costs of deindustrialization feed back into the democratic erosion that has characterized many advanced economies in recent decades.

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A Data Description

This section details the data that underpin the analysis. Summary statistics, plots, and data distributions are presented to provide a more rich understanding of the story and its suitability for research questions.

A.1 Distributional Change

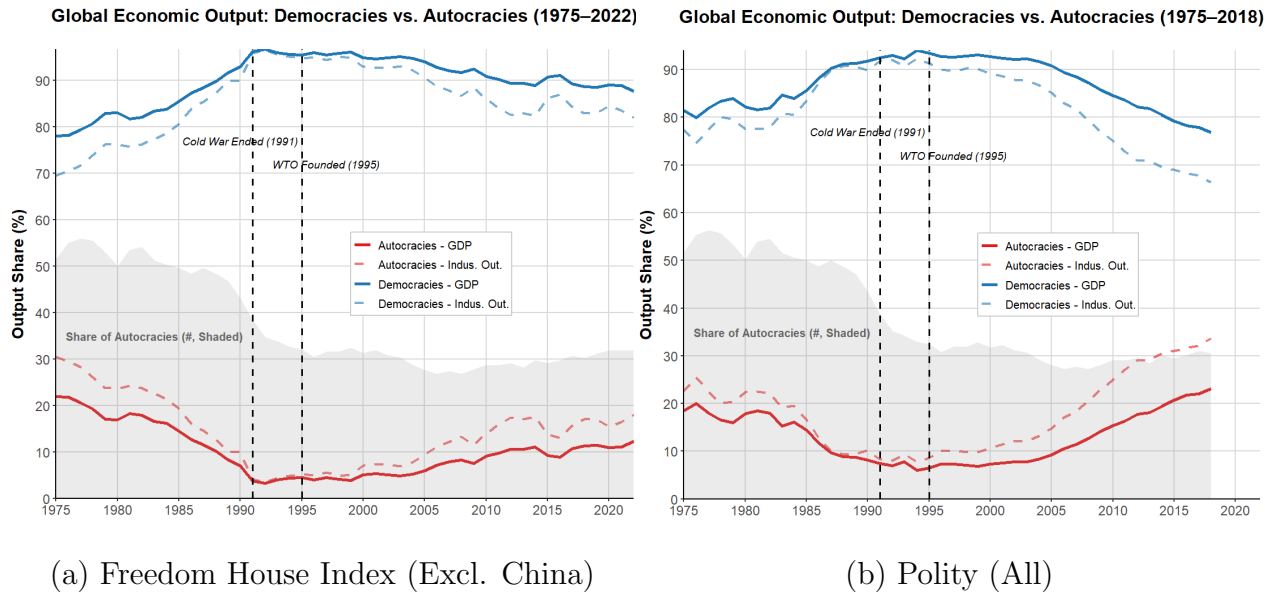


Figure A.1: The Distribution of Power Change Between Democracies and Autocracies. *Note:* Data: World Bank. In (a), autocracy is measured by $FH \geq 10$. In (b), autocracy is measured by $Polity \leq 0$. Polity data is available until 2018.

FIGURE 3: LIBERAL DEMOCRACY INDEX, GLOBAL AND REGIONAL AVERAGES 1971–2021

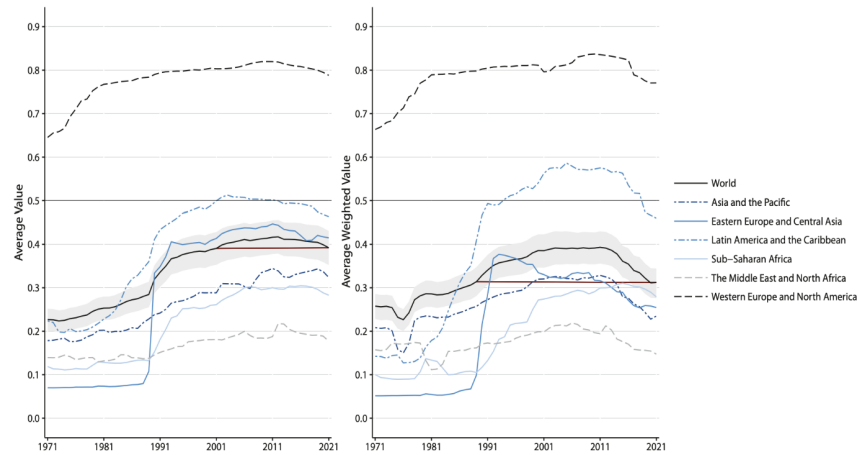


Figure A.2: VDem Global Liberal Democracy Index (Source: the Vdem Report 2021)

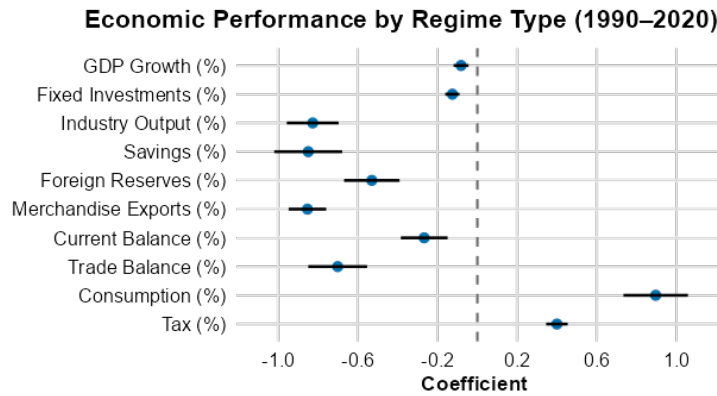


Figure A.3: Regime Type and Major Economic Indicators (World Bank). *Note:* The bar plot means the percentage point change associated with one unit increase in Polity, controlling for GDP per capita and year fixed-effects for similar development-level comparisons. Correlations are robust to excluding China or OPEC states and including continent dummies and natural resource rents (%).

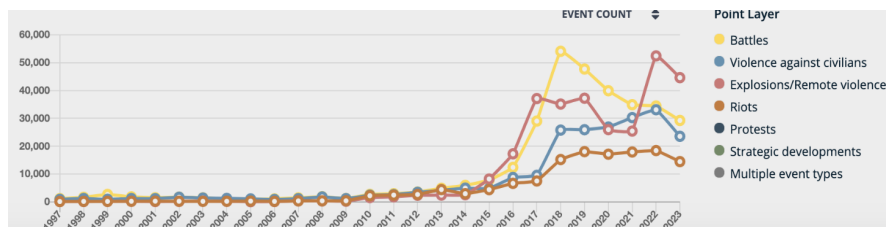


Figure A.4: The Trend of Global Conflicts (Source: ACLED).

Country/Leader	Fast Growth?	Export-Led?	Reason/Details
----------------	--------------	-------------	----------------

China (1978–2008)	Yes	Yes	Consistently high growth (10%), driven by market-oriented reform and export-led manufacturing, SEZs, and WTO entry.
Japan (1950s–1970s)	Yes	Yes	Post-war authoritarian, market-minded bureaucracy enabled export-led growth, tech upgrading, and 9% annual GDP growth.
South Korea (1962–1987)	Yes	Yes	Authoritarian state led industrial policy and export push.
Taiwan (1960s–1980s)	Yes	Yes	Under KMT authoritarianism, state-supported export sectors like electronics drove fast growth.
Singapore (1965–1990)	Yes	Yes	Semi-authoritarian regime used strategic export-oriented policy and MNC investment.
Hong Kong (1950s–1980s)	Yes	Yes	Colonial rule with modern market; booming re-export and service economy.
Malaysia (1986–2000)	Yes	Yes	Rapid growth post-1985 crisis; export-oriented industrialization (electronics, palm oil); strong FDI.
Thailand (1986–1996)	Yes	Yes	Export-led manufacturing boom before 1997 crisis; 9% annual growth driven by electronics and autos.
Indonesia (1989–1996)	Yes	Yes	Suharto-era liberalization brought export growth in textiles, natural resources before Asian financial crisis.
Vietnam (1990s–2010)	Yes	Yes	Post-Doi Moi reforms led to 6–7% growth; FDI-driven export economy in garments, electronics, and agri.
Pinochet (Chile, 1980s)	Yes	Yes	7% growth post-1985; market reform; exports quadrupled in 1985-1995
Franco (Spain, 1959–74)	Yes	Partial	average 7% growth rate; authoritarian technocrats with market reforms; GATT 1963, abandoned autarkic policies but on imports and low mercantilist policies; FDI, domestic industrialization mattered more than exports
Junta (Brazil, 1968-80)	Yes	Partial	10%+ growth rate during 'Brazilian Miracle'; market economy; state-led investment in infrastructure and industry and export promotion, exports increased ten-fold in 1968-1980
Salazar (Portugal, 1950/60s)	Yes	Yes	average 5–7% growth rate; market reform; investment in infrastructure and industry; export boom after 1960 EFTA and 1962 GATT, exports tripled in 1960-1970
PRI (Mexico, 1950s-80s)	Yes	Yes	1950-70 state-led industrialization, infra investment, and ISI; market institutions; FDI/tech absorption weaker than Spain; 1970s started export-oriented policies, exports increased thirteen-fold in 1972-82, 1986 GATT and 1994 NAFTA-led export boom.
Fujimori (Peru, 1993-2013)	Partial	Yes	Early ISI not work well; 1980/90s market reforms; post-Fujimori based on early reforms; riding on commodity boom, exports increased twelve-fold in 1993-2013.
dos Santos (Angola, 2002-12)	Yes	Yes	Market reform though limited, attracts foreign oil companies, state-led infrastructure; post-war boom largely driven by oil exports, exports increased by fourteen-fold; .
Ben Ali (Tunisia)	Yes	Yes	Growth via light manufacturing and exports to Europe, especially textiles.

Kagame (Rwanda)	Yes	Partial	7–8% annual growth, with modest export expansion and large public investment.
Botswana (1966–80)	Yes	Yes	Diamond mining accounted for most of Africa’s strongest GDP growth.
Zenawi (Ethiopia)	Yes	Partial	10% growth, largely from public investment, not heavily export-based.
Obiang (Eq. Guinea)	Yes	Yes	Explosive growth from oil, nearly all of GDP tied to exports.
Mubarak (Egypt)	Partial	Partial	Growth via gas, tourism, remittances, and modest export expansion.
Mohammed VI (Morocco)	Partial	Partial	Exports grew, but tourism and domestic consumption were also key.

Table A.1: Assessment of Growth and Export Strategy in Selected Autocratic Regimes. Note: I categorize “Fast Growth” and “Export-led” by Yes, Partial, and No.

A.2 Descriptive Performance

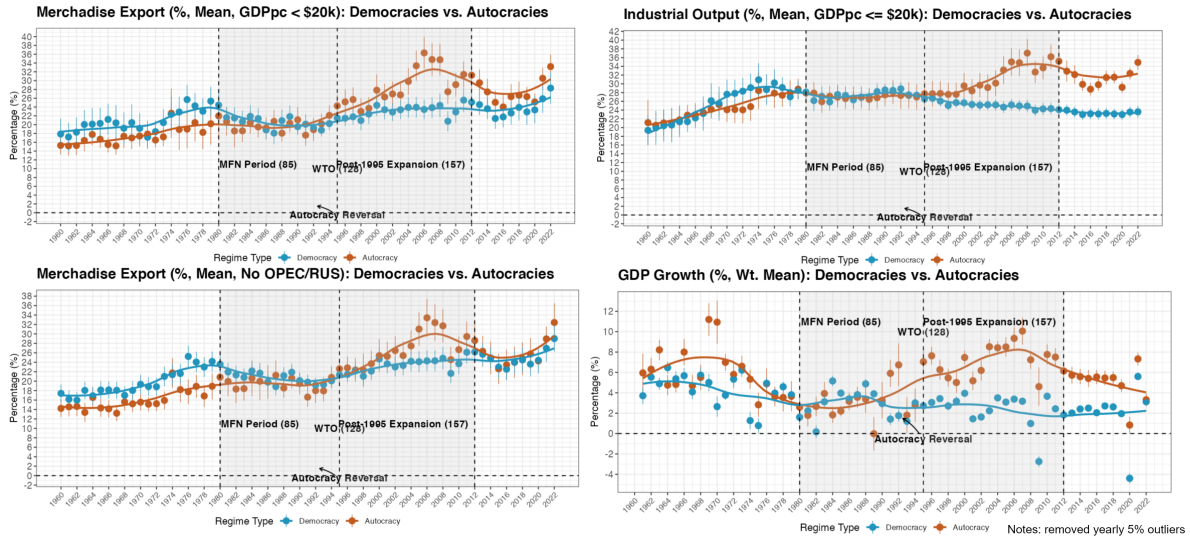
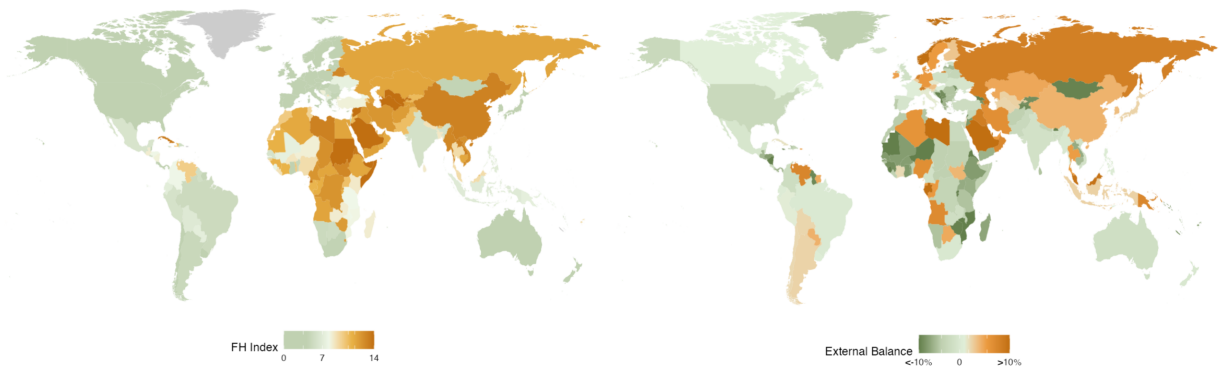


Figure A.5: Average Performance of Economic Indicators between Democracies and Autocracies (FH ≥ 11).



Freedom House Index

Average External Balance (% of GDP)

Figure A.6: Correlation between Regime Type and External Balance. *Note:* average external balance calculate the mean of current account balance and trade balance to include the information of both balances, since the two oftentimes do not overlap.

A.3 Panel Data of Regime Type Change

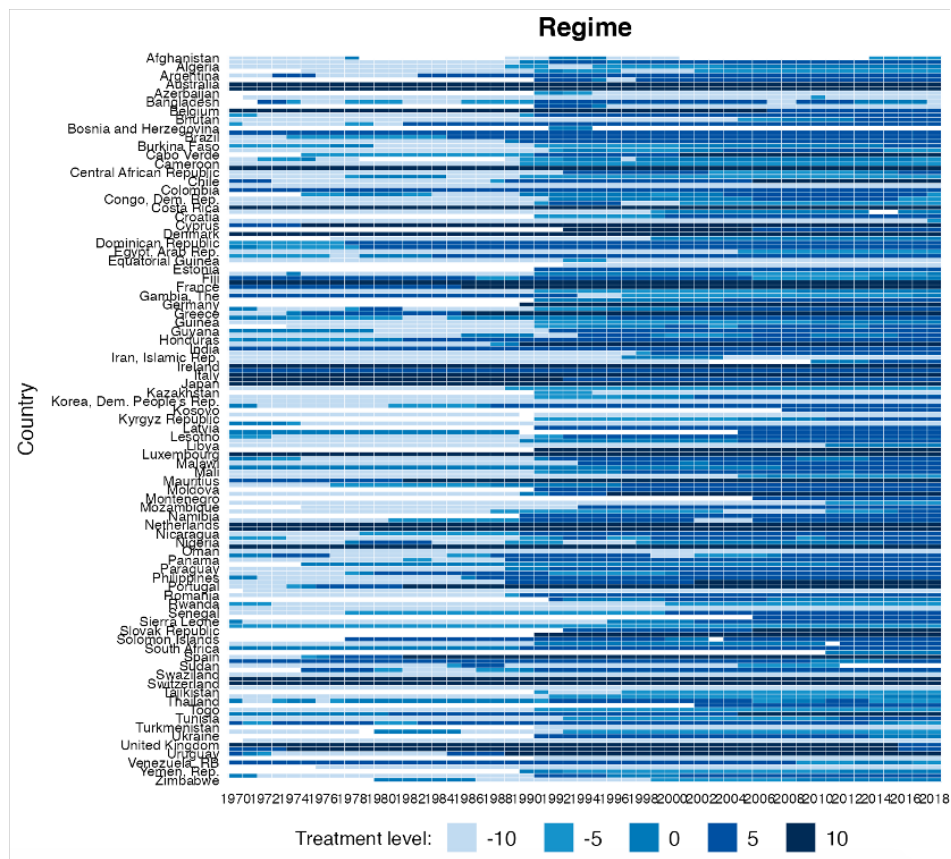


Figure A.7: Democratization (Polity Index)

A.4 KOF Globalization Index (Economic) Change

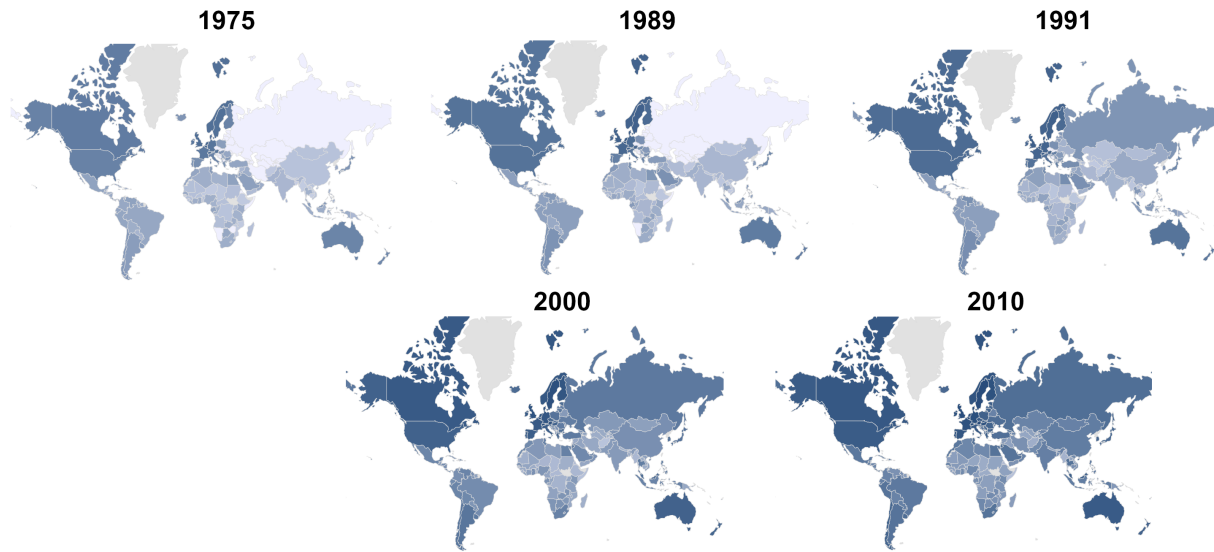


Figure A.8: KOF Globalization Index (Economic) Change. *Note:* The KOF economic index measures flows of trade, FDI and transfers, and trade and capital accounts restrictions.

A.5 WTO Membership

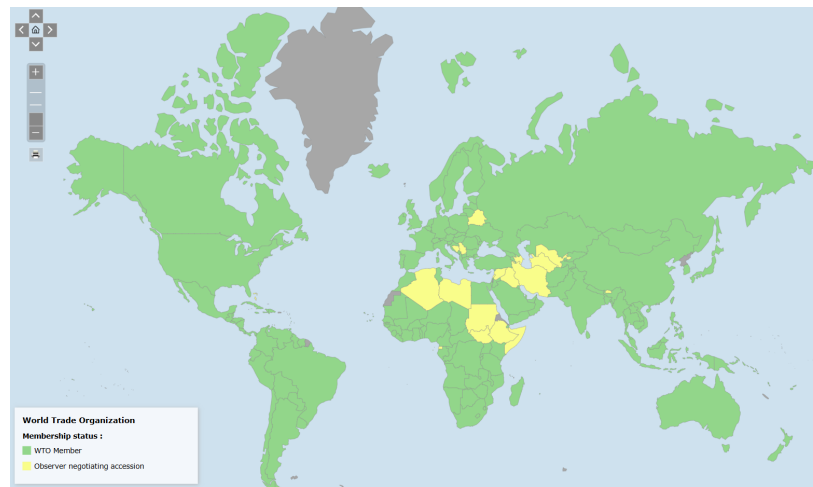


Figure A.9: The Map of WTO Members and Observers (source: WTO website)

B Theory Section

B.1 Formal Model

Part I: a Model of Engaged Reformers and Trade Manipulation

To formalize the theoretical logic, I present a model in which regimes acquire their levels of trade manipulation (vis-à-vis free trade practices), whether proactive or passive (e.g., industrial policy, wage suppression, or currency devaluation) to maintain political survival. The model combines leader perception, institutional structure, and the costs and benefits of manipulation, illustrating that autocratic engaged reformers are both more able and less costly to achieve a higher level of trade manipulation. The model formalizes the argument that the autocratic export advantage is conditional rather than automatic: it occurs when external engagement and basic market reform make export-oriented intervention effective.

Setup. Consider a regime choosing a level of export manipulation, $T \in [0, 1]$, which captures policies that improve export competitiveness, including industrial policy, directed credit, wage restraint, currency management, infrastructure support, and selective regulatory enforcement.

Regime type is denoted by $\phi \in \{0, 1\}$, where $\phi = 0$ denotes autocracy and $\phi = 1$ denotes democracy. Countries also differ in two institutional conditions. First, $w \in \{0, 1\}$ denotes external trade engagement, such as WTO-centered access to foreign markets. Second, $r \in [0, 1]$ denotes the degree of market-oriented reform.

Export manipulation is effective only when countries are externally engaged and sufficiently market-reformed. I capture this with an effectiveness coefficient function:

$$q(w, r) = w(\eta_0 + \eta_1 r - \eta_2 r^2), \tag{1}$$

where $\eta_0 \geq 0, \eta_1 > 0, \eta_2 > 0$. Thus, engagement is necessary for export manipulation to generate foreign-market returns. Reform initially increases the effectiveness of export

manipulation by improving market responsiveness, contract enforcement, and firm capacity. However, the effectiveness gains from reform are diminishing (hence r^2).

Regimes also differ in how much they value export performance. Let $\mu(\phi) > 0$ denote the political value of export returns. I assume $\mu(0) > \mu(1)$. This reflects the political survival logic: because autocracies have weaker domestic demand, less inclusive redistribution, and lower baseline domestic legitimacy, export performance carries greater political value for autocratic leaders.

The domestic institutional cost of export manipulation is:

$$\rho(\phi, r) = \rho_0 + \rho_\phi \phi + \rho_r r, \quad (2)$$

where $\rho_0 > 0, \rho_\phi > 0, \rho_r > 0$. The assumption $\rho_\phi > 0$ means democracies face higher domestic political costs from discretionary export manipulation because of electoral accountability, transparency, veto players, and legal constraints. The assumption $\rho_r > 0$ means deeper reform raises institutional constraints and limits arbitrary intervention—the costs.

Finally, the regime chooses T to maximize:

$$U(T; \phi, w, r) = \mu(\phi)\alpha q(w, r)T - \frac{1}{2}\rho(\phi, r)T^2 - \frac{1}{2}\gamma T^2, \quad (3)$$

where $\alpha > 0$ denotes baseline payoffs from manipulation. $\gamma > 0$ below captures international reputational, legal, or enforcement costs. T^2 implies the increasing cost of manipulation.

Equilibrium. The first-order condition is:

$$\frac{\partial U}{\partial T} = \mu(\phi)\alpha q(w, r) - [\rho(\phi, r) + \gamma]T = 0. \quad (4)$$

The interior solution is:

$$T^*(\phi, w, r) = \frac{\mu(\phi)\alpha q(w, r)}{\rho(\phi, r) + \gamma}. \quad (5)$$

Given $T \in [0, 1]$, the constrained equilibrium is:

$$T^*(\phi, w, r) = \min \left\{ 1, \max \left\{ 0, \frac{\mu(\phi)\alpha w(\eta_0 + \eta_1 r - \eta_2 r^2)}{\rho_0 + \rho_\phi \phi + \rho_r r + \gamma} \right\} \right\}. \quad (6)$$

Export manipulation is chosen only when $q(w, r) > 0$. Thus, neither autocracy nor reform alone is sufficient. Export manipulation requires external engagement and sufficient reform to generate foreign-market returns.

Proposition 1: Engagement is necessary for export manipulation. If $w = 0$, then $q(w, r) = 0$. Thus:

$$T^*(\phi, 0, r) = 0. \quad (7)$$

Without external engagement, export manipulation has no foreign-market payoff and is not chosen.

Proposition 2: Reform has a non-monotonic effect. When $w = 1$, the interior solution is:

$$T^*(\phi, 1, r) = \frac{\mu(\phi)\alpha(\eta_0 + \eta_1 r - \eta_2 r^2)}{\rho_0 + \rho_\phi \phi + \rho_r r + \gamma}. \quad (8)$$

Reform affects export manipulation through two channels. It increases effectiveness through $q(w, r)$, but it also raises domestic institutional costs through $\rho(\phi, r)$. Therefore:

$$\frac{\partial T^*}{\partial r} = \frac{\mu(\phi)\alpha \{(\eta_1 - 2\eta_2 r)[\rho(\phi, r) + \gamma] - q(1, r)\rho_r\}}{[\rho(\phi, r) + \gamma]^2}. \quad (9)$$

Dividing the numerator condition by $q(1, r)[\rho(\phi, r) + \gamma] > 0$, the sign of $\partial T^*/\partial r$ is positive

if and only if:

$$\frac{\eta_1 - 2\eta_2 r}{q(1, r)} > \frac{\rho_r}{\rho(\phi, r) + \gamma}, \quad (10)$$

where the left-hand side is the proportional marginal gain in effectiveness from reform and the right-hand side is the proportional marginal increase in costs from reform.

At low levels of reform, condition (10) holds. The left-hand side is large because $q(1, r) \approx \eta_0$ is small—underdeveloped markets generate low baseline effectiveness—while η_1 is large, as the first steps of reform yield high effectiveness gains. The right-hand side is small because $\rho(\phi, r) + \gamma$ is large relative to ρ_r , as institutional constraints have not yet accumulated. Thus $\partial T^*/\partial r > 0$.

At high levels of reform, condition (10) fails. As r increases, $\eta_1 - 2\eta_2 r$ falls and eventually turns negative, while $q(1, r)$ and $\rho_r/(\rho(\phi, r) + \gamma)$ remain positive. Thus $\partial T^*/\partial r < 0$.

Thus, export manipulation is strongest at intermediate levels of reform. Very low reform makes intervention ineffective; very high reform makes it institutionally costly.

Proposition 3: Conditional on engagement and reform, autocracies choose higher export manipulation. Under the maintained assumptions $\mu(0) > \mu(1)$ and $\rho(0, r) < \rho(1, r)$, autocracies choose higher export manipulation than democracies when the interior solution holds.

For autocracy:

$$T^*(0, w, r) = \frac{\mu(0)\alpha q(w, r)}{\rho(0, r) + \gamma}. \quad (11)$$

For democracy:

$$T^*(1, w, r) = \frac{\mu(1)\alpha q(w, r)}{\rho(1, r) + \gamma}. \quad (12)$$

Autocracies choose higher export manipulation when:

$$\frac{\mu(0)}{\rho(0, r) + \gamma} > \frac{\mu(1)}{\rho(1, r) + \gamma}. \quad (13)$$

This means that the condition is satisfied when autocracies attach sufficiently greater political value to export performance and face sufficiently lower domestic institutional costs.

Proposition 4: International enforcement reduces export manipulation. The effect of international enforcement costs is:

$$\frac{\partial T^*}{\partial \gamma} = -\frac{\mu(\phi)\alpha q(w, r)}{[\rho(\phi, r) + \gamma]^2} < 0. \quad (14)$$

Thus, stronger international enforcement reduces export manipulation.

The effect is larger for autocracies when:

$$\frac{\mu(0)}{[\rho(0, r) + \gamma]^2} > \frac{\mu(1)}{[\rho(1, r) + \gamma]^2}. \quad (15)$$

Thus, international enforcement reduces autocratic manipulation more strongly when autocracies rely sufficiently more on export performance and face sufficiently lower domestic institutional costs.

Implications. The model generates three empirical implications. First, autocracies should not systematically export more in general. Export advantage should appear primarily among autocracies that are externally engaged and sufficiently market-reformed. Second, reform should have a non-monotonic effect. Very low reform leaves export manipulation ineffective; very high reform raises institutional constraints. The strongest effect should appear among moderately reformed autocracies. Third, international enforcement should reduce discretionary export manipulation, especially among autocracies whose export strategy depends on policy discretion.

Part II: An Extended Eaton-Kortum Model

This section extends the classic Eaton-Kortum (E-K) model (Eaton and Kortum 2002) to illustrate the logic and hypotheses of this paper, since this model focuses on the determinants of bilateral trade flows. As in the paper, the trade flow from country i to j in E-K model is formally expressed as below:

$$X_{ij} = \frac{T_i(c_i\tau_{ij})^{-\theta}}{\sum_k T_k(c_k\tau_{kj})^{-\theta}} Y_j \quad (16)$$

where T_i is country i 's state of technology (or absolute advantage). c_i is cost of production, and τ_{ij} is trade barrier between two countries i and j . θ denotes the heterogeneity of a country's efficiency in producing a continuum of goods drawn from a Fréchet distribution ($F_i(z) = e^{-T_i z^{-\theta}}$) – that is, the comparative advantage. X_{ij} measures the trade flow from i to j , while Y_j is the aggregate consumption of country j . Together, (1) implies that a country's exports are determined by its technology, production cost, and trade cost.

In (2) - (4), three new variables related to my paper are introduced: institution (I_i), autocratic advantage (A_i , proxied by regime type), and WTO membership (W_i). In a globalized economy, trade-related productive technology (T_i) of country i is largely affected by investments, assuming more significantly from involving the GVC. Therefore, T_i depends on institutional improvement (I_i) and autocratic advantage (A_i) to attract the GVC, conditional on crossing an institutional threshold (I^*) and possessing WTO membership ($W_i = 1$, to proxy trade engagement). Both I^* and $W_i = 1$ are needed for substantively attracting firms, domestic or multinational, to invest and innovate. In other words, if institution is too low or excluded from the global trade system, institutional improvement or autocratic advantage won't matter much. Autocratic advantages A_i , embodied in state-market synergy,

magnify the effect of I_i . Technology function $t(A_i, I_i)$ can be thus formally written as:

$$t(A_i, I_i) = \begin{cases} \bar{T}_i \cdot \exp\left(\frac{\beta A_i}{1+\lambda A_i} + \gamma \frac{I_i A_i}{1+\lambda I_i^2}\right), & \text{if } I_i \geq I^* \text{ and } W_i = 1, \\ \bar{T}_i, & \text{otherwise.} \end{cases} \quad (17)$$

where $\bar{T}_i$ is the baseline technology. $\frac{\beta A_i}{1+\lambda A_i}$ models diminishing returns to A_i , while $\lambda > 0$ controls how quickly diminishing returns to A_i sets in. When A_i is small: $\frac{\beta A_i}{1+\lambda A_i} \approx \beta A_i$, so the contribution of A_i grows almost linearly. When A_i is large: $\frac{\beta A_i}{1+\lambda A_i} \rightarrow \frac{\beta}{\lambda}$, so the marginal effect of A_i diminishes significantly and possibly goes negative when A_i too high. Similarly, $\frac{I_i A_i}{1+\lambda I_i^2}$ models diminishing returns to I_i . The economic rationale behind is that excessive centralization (high A_i) may create inefficiencies or governance rigidities, while saturated institutions (high I_i) may over-complicate decision-making, reducing efficiency. A moderately autocratic regime (e.g., with some centralized control) may gain substantial benefits, but extreme autocracy may lead to inefficiencies (e.g., power abuse).

Although I_i and A_i exhibit diminishing returns, they may still interact to amplify technology – i.e., the interaction term $\gamma \frac{I_i A_i}{1+\lambda I_i^2}$. γ captures the interaction coefficient between I_i and A_i on (T_i) . Moderate levels of I_i and A_i together create the largest gains in technology because they complement each other. For example, moderate autocratic regimes (e.g., single-party regimes) may gain disproportionately when combined with moderate to high institutions. $\frac{I_i A_i}{1+\lambda I_i^2}$ also captures that when I_i is high, the effect of A_i diminishes (the denominator is dominated by I_i^2), because even autocratic regimes are now more hands-tied, if not completely disabled.

As argued, production cost c_i decreases with autocratic advantage A_i , due to reasons such as labor rights suppression and disproportionate state support for industries and infrastructure. c_i is formally expressed as:

$$c(A_i) = \begin{cases} \bar{c}_i \cdot \exp(-\delta A_i), & \text{if } I_i \geq I^* \text{ and } W_i = 1, \\ \bar{c}_i, & \text{otherwise.} \end{cases} \quad (18)$$

where $\bar{c}_i$ is the baseline production cost and δ_A captures the cost-reducing effect of autocratic advantage. The term $\exp(-\delta_A A_i)$ represents an exponential decay. As A_i increases, $\exp(-\delta_A A_i)$ becomes smaller, but the rate of decrease slows down because the exponential decay flattens over A_i . The intuition is that there's a limit to how much A_i can reduce costs, as well as marginal diminishing returns. The effective productive cost reduction is also assumed to be conditioned by crossing certain I^* and $W_i = 1$, otherwise, A_i may have limited effect.

Trade cost τ_{ij} decreases with WTO membership W_i , formally as:

$$\tau(W_i) = \bar{\tau}_{ij} \cdot (1 - \varphi W_i) \quad (19)$$

where τ_{ij} is baseline trade cost, and $\lambda_W > 0$ indicates reduction in trade costs due to WTO membership (or trade engagement). Therefore, (1) takes on the form of:

$$X_{ij} = \frac{t(A_i, I_i) \{c(A_i) \tau(W_i)\}^{-\theta}}{\sum_k t(A_k, I_k) \{c(A_k) \tau(W_k)\}^{-\theta}} Y_j \quad (20)$$

Plug (2)-(4) into (5), the full trade flow formula becomes:

$$X_{ij} = \frac{\left[\bar{T}_i \cdot \exp\left(\frac{\beta A_i}{1+\lambda A_i} + \gamma \frac{I_i A_i}{1+\lambda I_i^2}\right) \right] \cdot \{[\bar{c}_i \cdot \exp(-\delta A_i)] \cdot [\bar{\tau}_{ij} \cdot (1 - \varphi W_i)]\}^{-\theta}}{\sum_k T_k \cdot (c_k \tau_{kj})^{-\theta}} \cdot Y_j, \quad \text{if } I_i \geq I^* \text{ and } W_i = 1. \quad (21)$$

From (5), it shows that if $W_i = 0$, meaning a country i is not engaged in the global trade system, trade flow from i to j is simplified to the original baseline form (6), which means neither institutional improvement nor increased autocratic advantages (proxied by more autocratic regime type) will significantly improve trade flows.

$$X_{ij} = \frac{\bar{T}_i (\bar{c}_i \bar{\tau}_{ij})^{-\theta}}{\sum_k T_k \cdot (c_k \tau_{kj})^{-\theta}} \cdot Y_j \quad (22)$$

When W_i changes from 0 to 1, trade cost decreases, so that trade flow from X_{ij} increases.

An increase in autocratic advantages A_i leads to an increase in productive technology:

$$\bar{T}_i \cdot \exp \left(\frac{\beta A_i}{1 + \lambda A_i} + \gamma \frac{I_i A_i}{1 + \lambda I_i^2 + \varepsilon A_i} \right)$$

and a decrease in productive cost $\bar{c}_i \cdot \exp(-\delta A_i)$. Therefore, exports X_{ij} increases. This is consistent with *H1.1*.

Note that in order for productive technology to increase and productive cost to decrease, I_i has to cross certain thresholds (i.e., $I_i > I^*$), and it cannot be too high. This comes from conditions in (2) and (3) and is consistent with *H1.2*. To combine all, when W_i changes from 0 to 1 and $I_i > I^*$, (5) minus (6) becomes:

$$\Delta X_{ij} = \frac{\left[\bar{T}_i \cdot \exp \left(\frac{\beta A_i}{1 + \lambda A_i} + \gamma \frac{I_i A_i}{1 + \lambda I_i^2} \right) \right] \cdot \{ [\bar{c}_i \cdot \exp(-\delta A_i)] \cdot [\bar{\tau}_{ij} \cdot (1 - \varphi)] \}^{-\theta} - \bar{T}_i (\bar{c}_i \bar{\tau}_{ij})^{-\theta}}{\sum_k T_k \cdot (c_k \tau_{kj})^{-\theta}} \cdot Y_j \quad (23)$$

(7) formally denotes that after passing institutional thresholds I^* , an increase in A_i leads to more export increase for the same WTO accession. Put differently, autocratic WTO-joiners are expected to experience more export increase than their democratic counterparts.

Similarly, when $W_i = 0$, an increase in institution will not increase $\bar{T}_i$ or decrease $\bar{c}_i$, thus not increasing X_{ij} , as specified by (2) and (3). However, when $W_i = 1$, for a ΔI increase in institutional level while $\bar{c}_i \cdot \exp(-\delta A_i)$ and $\bar{\tau}_{ij} \cdot (1 - \varphi W_i)$ keep unchanged, change in X_{ij} is expressed and simplified as:

$$\Delta X_{ij} = \frac{\left[\bar{T}_i \cdot \exp \left(\frac{\beta A_i}{1 + \lambda A_i} \right) \exp \left(\frac{I_i + \Delta I}{1 + \lambda (I_i + \Delta I)^2} - \frac{I_i}{1 + \lambda I_i^2} \right) \right] \cdot \{ [\bar{c}_i \cdot \exp(-\delta A_i)] \cdot [\bar{\tau}_{ij} \cdot (1 - \varphi)] \}^{-\theta}}{\sum_k T_k \cdot (c_k \tau_{kj})^{-\theta}} \cdot Y_j \quad (24)$$

(8) also implies an increase in A_i leads to more export increase for the same institutional improvement, given crossing institutional thresholds and trade engagement. In other words, more autocratic states engaged by the global trade regime should expect more gains from institutional improvement in a globalized economy. Again, A_i may be subject to possible

decreasing marginal returns.

Last, since X_{ij} denotes the absolute level of trade flow from i to j , (5) also implies that more autocratic states (larger A_i) are expected to “inflate” trade flows more conditional on other factors such as trade engagement and institutional levels. However, this is also subject to the conditions in (2) and (3). For example, for states that all have $W_i = 1$, A_i ’s effect may diminish when I_i doesn’t cross thresholds I^* or is too high. For states that all have similarly moderate institutional levels I_i , A_i ’s effect may diminish when $W_i = 0$.

B.2 Predictive Patterns

Prediction on Exports

$$X_{ijk} = \frac{s_{ik}Y_iY_j}{(p_{ik})^\sigma \bar{y}_{ik}} \left[T_{ijk}(z_i, z_j) / P_j^k \right]^{1-\sigma} [\theta_{ik} \exp(z_i)]^{\sigma-1}$$

Example of Gravity Model Incorporating Product Quality, Yu(2010)

First, similar to Yu (2010), by employing gravity model commonly used in economics and political science (Anderson and van Wincoop 2003) controlling for a standard list of dyad-level covariates, I find that prior to 1990, being more democratic is associated with higher exports (see Table B.2). Post-1990, however, being more autocratic is associated with a positive or zero effect compared to being more democratic is.⁵² The models include cross-sectional, within-exporter, interaction with exporter’s logged GDP (whether the coefficient differs for larger countries), and weighted least squares (when larger countries are assigned larger weights). Using the interaction model, for example, by plugging in Iran’s GDP in 2005 (the logged form = 20), the effect of Polity is negative.

⁵²For post-1990, I look at all dyads with exporter being within the WTO, since many autocracies joined the WTO after 1990 and being inside the WTO is what I am interested in. In contrast, the pre-1990 model checks both inside and outside of the WTO since most autocracies were excluded. However, the result barely changes if WTO only.

	Pre-1990	Post-1990 (within WTO)			
	OLS	OLS	OLS (Within)	OLS (Interaction)	WLS (by GDP)
<i>Polity_i</i>	0.022*** (0.001)	0.003 (0.003)	-0.016*** (0.004)	0.065*** (0.014)	-0.041*** (0.001)
<i>Polity_i</i> x <i>GDP_i</i>				-0.004*** (0.001)	
<i>Polity_j</i>	0.003* (0.001)	0.003** (0.001)	0.005*** (0.001)	0.003** (0.001)	0.003*** (0.001)
<i>GDP_i</i>	1.583*** (0.048)	-1.746*** (0.139)	0.298*** (0.054)	-1.698*** (0.141)	-1.137*** (0.107)
<i>GDP_j</i>	2.058*** (0.143)	0.650*** (0.080)	0.521*** (0.090)	0.644*** (0.079)	0.065 (0.182)
<i>GDPPC_i</i>	-0.536*** (0.043)	3.051*** (0.129)	0.070 (0.063)	3.023*** (0.129)	2.154*** (0.116)
<i>GDPPC_j</i>	-1.011*** (0.137)	0.387*** (0.083)	0.526*** (0.092)	0.392*** (0.082)	0.967*** (0.185)
RTA	0.204*** (0.051)	0.282*** (0.041)	0.286*** (0.039)	0.278*** (0.041)	0.202*** (0.031)
Custom Union	0.819*** (0.111)	0.590*** (0.031)	0.662*** (0.032)	0.591*** (0.030)	-0.335*** (0.030)
Common Colnizer post-45	0.775*** (0.027)	0.998*** (0.022)	0.875*** (0.017)	0.996*** (0.022)	0.533*** (0.047)
Colonial Dep. post-45	1.724*** (0.044)	1.034*** (0.061)	1.260*** (0.048)	1.048*** (0.060)	0.982*** (0.050)
<i>Population_i</i>	-0.757*** (0.041)	2.984*** (0.124)	-0.112 (0.093)	2.955*** (0.124)	2.190*** (0.109)
<i>Population_j</i>	-1.195*** (0.139)	0.445*** (0.072)	0.596*** (0.082)	0.450*** (0.071)	0.938*** (0.180)
Distance	-0.785*** (0.025)	-1.106*** (0.011)	-1.244*** (0.015)	-1.109*** (0.011)	-0.960*** (0.013)
Common Language	0.294*** (0.023)	0.607*** (0.026)	0.685*** (0.030)	0.612*** (0.026)	0.331*** (0.020)
Common Religion	-0.059* (0.030)	-0.008 (0.032)	0.266*** (0.033)	-0.010 (0.032)	-0.025 (0.028)
Border	0.462*** (0.017)	0.806*** (0.023)	0.529*** (0.023)	0.802*** (0.022)	0.073 (0.044)
Num.Obs.	194 716	313 566	313 566	313 566	313 580
R2 Adj.	0.629	0.709	0.759	0.709	0.801
FE	year	year	year/exporter	year	year

* p < 0.1, ** p < 0.05, *** p < 0.01

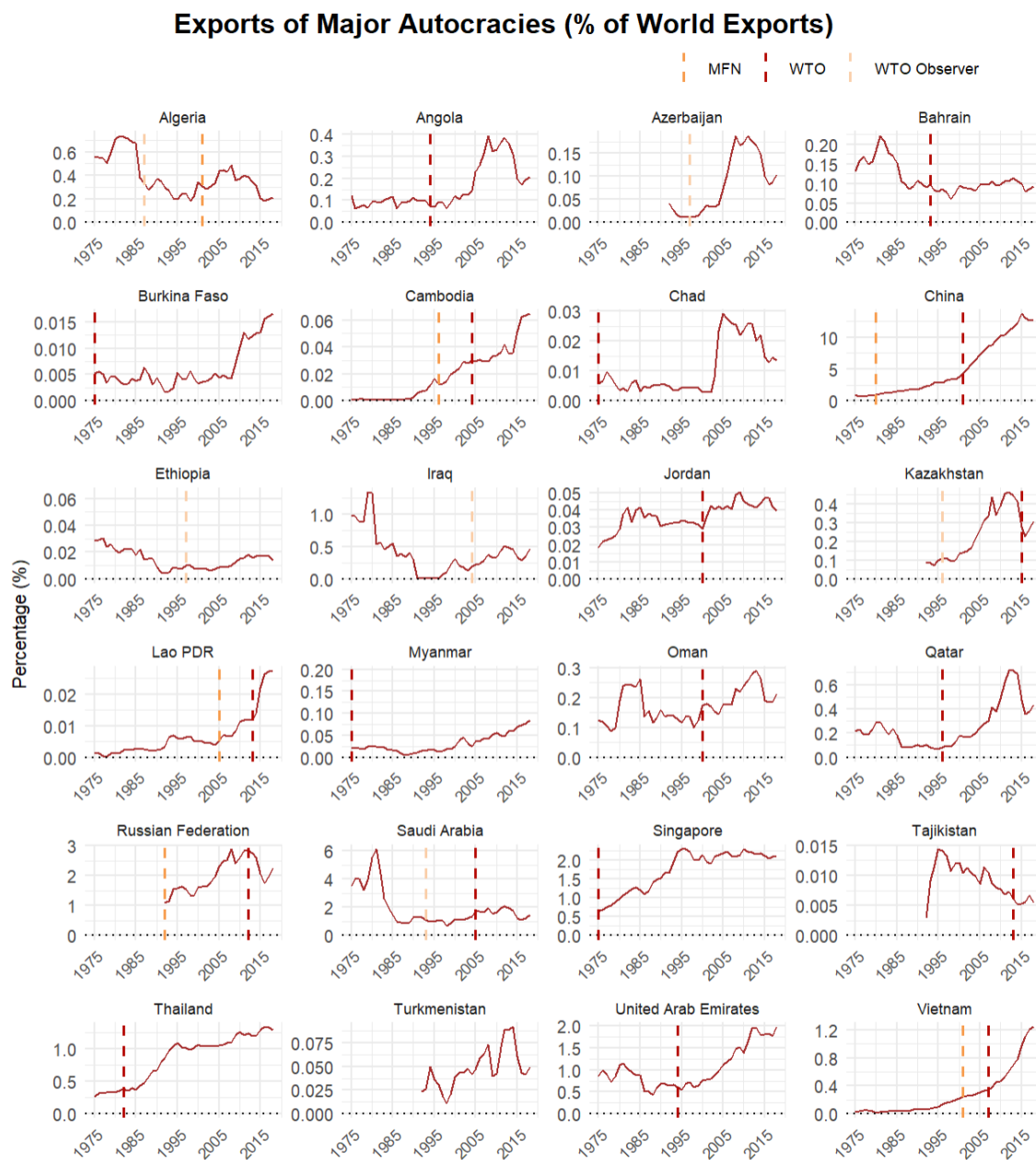
Table B.2: Regime Type and Exports

B.3 Selection of Institutional Thresholds

In 2000, the bottom 20 percentile threshold is 0.45 for PR protection and 0.2 for rule of law, respectively. I combine the institutional levels at the bottom 20 percentile among de-

veloping countries in 2000 and real cases (e.g., China's PR protection is around 0.35), so the thresholds are roughly 0.2 for rule of law and 0.35 for PR protection. Both values have to be reached. However, special cases remain. First, I slightly prioritize PR protection especially for resource-rich countries, for it is more attractive to the GVC than rule of law – as long as global investor's property rights are protected, global firms may more rely on within-GVC contract enforcement. For example, Cameroon and Chad, two resource-rich African countries have high PR protection (0.8 and 0.78) but low rule of law (below 0.1), for which I classify them as reformed. Azerbaijan (0.61, .03) and Equatorial Guinea (0.45, 0.06) are two other cases. Second, I factor in expectation. Venezuela's values for two indicators were 0.58 and 0.05 in 2010. Yet, Venezuela has experienced rapid institutional deterioration since 1997 before Hugo Chávez was elected completely reversing course when the two indicators were as high as 0.9 and 0.55, generating greatly adverse expectations for investors. Thus, Venezuela is listed as non-reformer. Yemen is another example: from the Arab Spring in 2011 to Houthi's takeover in 2015, its institutions experienced rapid deterioration.

B.4 Share of World Exports for Major Autocracies



Source: World Bank

Table B.3: Share of World Exports for Major Autocracies. *Note:* For illustration purpose, vertical dashed lines begin in 1975 if MFN/WTO/Observer in effect earlier. Most autocracies' global share in exports show an increasing trend.

C Main Empirical Tests

C.1 Main test

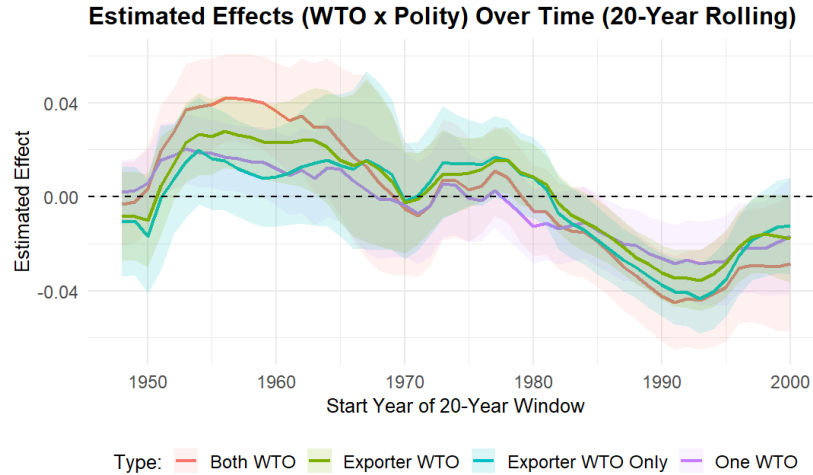


Figure C.10: Rolling Estimates of Various Types of GATT/WTO Effects Moderated by Polity. *Note:* The effects are estimated by gravity models above within 20-year rolling windows from 1948 to 2020. For example, 1980 denotes the window 1980-2000.

C.2 Sensitivity Analysis

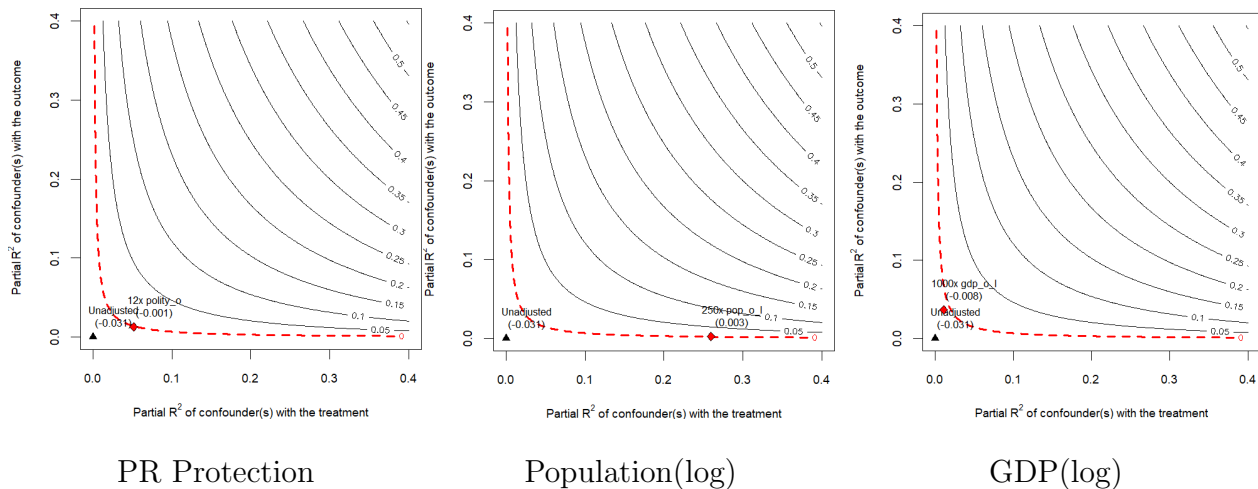


Figure C.11: Improved Covariate Balance via CBPS Weighting, Post-1990. *Note:* overall, all covariates' balance improve significantly. The green line natural resource intensity is slightly not balanced. However, it may not be an concern as it does not significantly affect WTO accession theoretically.

C.3 WTO Effect

	DV: 1. Exports (FE)		2. Exports (FE)		3. Exports (FE)		4. Imports (FE)		5. Exports (CRE)	
	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990
<i>One WTO</i>	0.304*** (0.052)	-0.129* (0.061)								
<i>One WTO</i> × <i>Polity_i</i>	-0.003 (0.005)	-0.025** (0.008)								
<i>Both WTO</i>	0.615*** (0.084)	0.142 (0.092)	0.314*** (0.046)	0.271*** (0.047)					0.309*** (0.015)	0.208*** (0.012)
<i>Both WTO</i> × <i>Polity_i</i>	0.007 (0.006)	-0.041*** (0.010)	0.011** (0.004)	-0.033*** (0.006)					0.012*** (0.001)	-0.038*** (0.002)
<i>WTO_i Only</i>			0.306*** (0.052)	-0.036 (0.065)					0.304*** (0.017)	0.099*** (0.017)
<i>WTO_i Only</i> × <i>Polity_i</i>			0.005 (0.005)	-0.037*** (0.007)					0.004* (0.002)	-0.038*** (0.002)
<i>WTO_i</i>					0.307*** (0.042)	0.208*** (0.045)	0.050 (0.038)	0.053 (0.047)		
<i>WTO_i</i> × <i>Polity_i</i>					0.009* (0.004)	-0.033*** (0.006)	0.000 (0.003)	-0.011 (0.006)		
<i>GDP_i</i>	-0.086 (0.085)	0.216* (0.090)	-0.087 (0.085)	0.212* (0.090)	-0.097 (0.085)	0.203* (0.089)	0.882*** (0.073)	0.780*** (0.064)	-0.022 (0.025)	0.273*** (0.019)
<i>GDP</i> PC _i	0.659*** (0.080)	0.276** (0.092)	0.659*** (0.080)	0.283** (0.092)	0.671*** (0.079)	0.285** (0.092)	-0.118 (0.066)	-0.050 (0.070)	0.607*** (0.025)	0.211*** (0.019)
<i>PTA</i>	0.200*** (0.032)	0.190*** (0.030)	0.200*** (0.032)	0.192*** (0.030)	0.201*** (0.032)	0.192*** (0.030)	0.192*** (0.032)	0.044 (0.029)	0.149*** (0.012)	0.191*** (0.013)
<i>RTA</i>	0.103* (0.045)	0.024 (0.031)	0.103* (0.045)	0.024 (0.031)	0.106* (0.045)	0.020 (0.031)	0.195*** (0.042)	-0.006 (0.029)	0.118*** (0.021)	0.006 (0.015)
<i>Customs Union</i>	0.249 (0.143)	0.088 (0.083)	0.250 (0.144)	0.096 (0.082)	0.249 (0.144)	0.099 (0.083)	0.230 (0.121)	0.174* (0.083)	0.342*** (0.049)	0.131*** (0.034)
<i>Colonial Dependency</i>	0.628*** (0.087)	1.432*** (0.216)	0.628*** (0.087)	1.437*** (0.216)	0.629*** (0.088)	1.442*** (0.216)	0.268** (0.099)	1.500*** (0.264)	0.630*** (0.118)	1.220** (0.406)
<i>Polity_i</i>	0.013* (0.005)		0.009* (0.004)		0.009* (0.004)		0.013*** (0.003)		0.009*** (0.001)	
<i>Com. Col. Post45</i>									0.443*** (0.056)	0.734*** (0.042)
<i>Col. Dep. Post45</i>									2.198*** (0.136)	1.731*** (0.125)
<i>Distance</i>									-1.079*** (0.020)	-1.380*** (0.016)
<i>Common Language</i>									0.213*** (0.045)	0.704*** (0.036)
<i>Common Religion</i>									0.084 (0.063)	0.326*** (0.048)
<i>Bordered</i>									0.598*** (0.090)	0.860*** (0.080)
Exporter Means									✓	✓
Dyad Means									✓	✓
Importer-year FE	✓	✓	✓	✓	✓	✓	✓	✓		
Dyad FE	✓	✓	✓	✓	✓	✓	✓	✓	RE	RE
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Num.Obs.	220 706	540 324	220 706	540 324	220 706	540 324	215 680	530 191	216 643	523 818
R2	0.879	0.893	0.879	0.893	0.879	0.893	0.878	0.887		
R2 Adj.	0.868	0.886	0.868	0.886	0.868	0.886	0.866	0.879		
R2 Within Adj.	0.035	0.012	0.035	0.012	0.035	0.012	0.042	0.021		
BIC	866 976.4	2 262 286.9	866 969.8	2 262 074.3	866 966.1	2 262 265.9	855 684.9	2 256 659.6	713 737.5	1 907 045.7

* p < 0.05, ** p < 0.01, *** p < 0.001

Table C.4: The Effects of Joining the WTO

	Exports (FE Model)		Exports (CRE Model)	
	PR Protection	RoL	PR Protection	RoL
<i>WTO_i : polity2000</i>	0.033 (0.040)	0.165*** (0.036)	-0.068*** (0.015)	0.122*** (0.019)
<i>WTO_i : polity2000 : inst_prpty_low</i>	-0.057 (0.046)		0.044** (0.017)	
<i>WTO_i : polity2000 : inst_prpty_mid</i>	-0.092** (0.041)		0.009 (0.015)	
<i>WTO_i : polity2000 : inst_rule_low</i>		-0.083** (0.040)		-0.033* (0.020)
<i>WTO_i : polity2000 : inst_rule_mid</i>		-0.226*** (0.037)		-0.183*** (0.019)

<i>WTO_i</i>	-0.365 (0.285)	-1.265*** (0.307)	0.270** (0.112)	-1.020*** (0.152)
<i>Polity_i</i>	-0.025*** (0.004)	-0.025*** (0.004)	-0.024*** (0.001)	-0.024*** (0.001)
<i>Polity_j</i>	-0.001 (0.003)	-0.001 (0.003)	0.004*** (0.001)	0.003*** (0.001)
<i>WTO_j</i>	-0.055 (0.064)	-0.059 (0.063)	-0.110*** (0.030)	-0.107*** (0.030)
<i>Both WTO</i>	0.217*** (0.065)	0.220*** (0.064)	0.268*** (0.031)	0.266*** (0.031)
<i>GDP_i</i>	1.770 (16.391)	1.603 (16.287)	-0.026 (6.526)	-0.237 (6.519)
<i>GDP_j</i>	1.304*** (0.243)	1.298*** (0.244)	1.393*** (0.171)	1.386*** (0.170)
<i>GDPPC_i</i>	-1.339 (16.391)	-1.192 (16.288)	0.455 (6.525)	0.651 (6.519)
<i>GDPPC_j</i>	-0.568** (0.242)	-0.562** (0.244)	-0.513*** (0.171)	-0.507*** (0.170)
<i>PTA</i>	0.117*** (0.038)	0.111*** (0.038)	0.146*** (0.019)	0.142*** (0.019)
<i>RTA</i>	0.067* (0.040)	0.061 (0.039)	0.088*** (0.021)	0.083*** (0.021)
<i>Customs Union</i>	-0.008 (0.097)	-0.019 (0.097)	0.067 (0.044)	0.058 (0.044)
<i>Population_i</i>	-1.673 (16.390)	-1.509 (16.286)	0.226 (6.526)	0.419 (6.519)
<i>Population_j</i>	-0.177 (0.250)	-0.192 (0.251)	-0.325* (0.171)	-0.321* (0.170)
<i>WTO_i : inst_prpty_low</i>	0.086 (0.302)		-0.612*** (0.124)	
<i>WTO_i : inst_prpty_mid</i>	0.377 (0.288)		-0.308*** (0.111)	
<i>WTO_i : inst_rule_low</i>		0.921*** (0.325)		0.630*** (0.152)
<i>WTO_i : inst_rule_mid</i>		1.325*** (0.302)		1.041*** (0.150)
<i>polity2000</i>			0.152 (0.141)	-0.058 (0.097)
<i>inst_prpty_catlow</i>			-0.266 (1.192)	
<i>inst_prpty_catmid</i>			0.165 (1.156)	
<i>Common Colonizer</i>			0.789*** (0.057)	0.785*** (0.057)
<i>Colonial Dep.</i>			2.072*** (0.191)	2.077*** (0.191)
<i>Distance</i>			-1.437*** (0.023)	-1.440*** (0.023)
<i>Common Language</i>			0.687*** (0.051)	0.687*** (0.051)
<i>Common Religion</i>			0.271*** (0.068)	0.273*** (0.068)
<i>Bordered</i>			0.911*** (0.106)	0.912*** (0.106)
<i>polity2000 : inst_prpty_catlow</i>			-0.043 (0.160)	
<i>polity2000 : inst_prpty_catmid</i>			-0.099 (0.148)	

<i>inst_rule_catlow</i>				-1.207
				(0.974)
<i>inst_rule_catmid</i>				-0.809
				(0.822)
<i>polity2000 : inst_rule_catlow</i>				0.066
				(0.177)
<i>polity2000 : inst_rule_catmid</i>				0.167
				(0.104)
Exporter Means			✓	✓
Dyad Means			✓	✓
Exporter FE	✓	✓	RE	RE
Importer FE	✓	✓		
Dyad FE	✓	✓	RE	RE
Year FE	✓	✓	✓	✓
Num.Obs.	284 345	284 345	275 914	275 914
R2	0.866	0.866		
R2 Adj.	0.858	0.858		
R2 Marg.			0.420	0.400
R2 Cond.			0.860	0.860
BIC	1 201 319.8	1 200 671.1	1 034 876.7	1 034 368.6

* p < 0.1, ** p < 0.05, *** p < 0.01

Table C.5: The Effects of Joining the WTO by Institutional Range.

	DV: Current Account Balance (%)						80-95
	FE	FE	RE	RE/No OPEC	RE/Developing	RE/WTO Post-90	
<i>Polity</i>	-0.212*** (0.039)	-0.123*** (0.043)	-0.146*** (0.049)	-0.135*** (0.049)	-0.139*** (0.053)	-0.225* (0.119)	0.095* (0.052)
<i>GDP (log)</i>		1.744*** (0.129)	1.430*** (0.265)	1.427*** (0.271)	1.440*** (0.296)	1.032 (0.707)	0.433 (0.414)
<i>GDPPC (log)</i>		0.787*** (0.251)	-0.148 (0.399)	-0.624 (0.415)	-0.385 (0.449)	1.229 (0.916)	-0.121 (0.697)
<i>GDP Growth</i>		-0.182** (0.093)	-0.118*** (0.031)	-0.146*** (0.032)	-0.127*** (0.035)	-0.034 (0.063)	-0.074 (0.046)
<i>Net Borrowing(%)</i>		0.657*** (0.061)	0.475*** (0.027)	0.453*** (0.031)	0.519*** (0.032)	0.657*** (0.053)	-0.014 (0.072)
<i>Foreign Asset(%)</i>		0.418* (0.231)	0.068 (0.119)	0.060 (0.114)	0.053 (0.128)	-0.065 (0.158)	0.724** (0.301)
<i>KA Open</i>		-0.264* (0.140)	-0.342** (0.172)	-0.230 (0.171)	-0.382** (0.185)	-0.370 (0.390)	-0.382 (0.287)
<i>ΔPrivate Credit (%)</i>		-0.195*** (0.038)	-0.139*** (0.015)	-0.138*** (0.014)	-0.178*** (0.021)	-0.238*** (0.041)	-0.127*** (0.036)
<i>ΔTerm of Trade</i>		0.064** (0.026)	0.085*** (0.011)	0.061*** (0.012)	0.096*** (0.012)	0.082*** (0.019)	0.031** (0.015)
<i>Population (under 14, %)</i>		15.831*** (4.223)	23.902*** (5.444)	19.920*** (5.558)	25.587*** (6.330)	55.684*** (11.228)	13.231 (11.693)
<i>Population (over 65, %)</i>		5.094 (6.103)	41.690*** (8.405)	41.891*** (8.305)	49.794*** (11.467)	98.099*** (21.183)	42.435 (34.278)
<i>Trade Openness</i>		0.041*** (0.005)	0.041*** (0.006)	0.043*** (0.006)	0.028*** (0.008)	-0.013 (0.015)	-0.004 (0.018)
<i>Year</i>		-0.316***	-0.278***	0.024	-0.039	-0.118	0.299***

		(0.062)	(0.082)	(0.165)	(0.048)	(0.112)	(0.070)
Year FE	✓	✓	✓	✓	✓	✓	✓
Country RE			✓	✓	✓	✓	✓
Num.Obs.	2704	1499	1499	1698	1469	540	290
R2 Marg.			0.422	0.384	0.322	0.420	0.252
R2 Cond.			0.759	0.757	0.691	0.755	0.605
BIC	43 454.6	20 230.6					

* p < 0.1, ** p < 0.05, *** p < 0.01

Table C.6: Regime Type's Effect on Current Account Balance.

	DV: Trade Balance (%)						
	FE	FE	RE	RE/No OPEC	RE/Developing	RE/WTO Post-90	80-95
Polity	−0.410*** (0.050)	−0.480*** (0.069)	−0.241*** (0.056)	−0.179*** (0.055)	−0.227*** (0.064)	−0.268** (0.124)	0.140** (0.057)
<i>GDP (log)</i>		2.594*** (0.183)	1.692*** (0.453)	1.896*** (0.442)	2.189*** (0.534)	0.309 (1.587)	0.411 (0.845)
<i>GDPPC (log)</i>		4.872*** (0.411)	3.032*** (0.592)	2.002*** (0.595)	3.007*** (0.698)	7.007*** (1.703)	−2.969*** (1.142)
<i>GDP Growth</i>		0.054 (0.102)	−0.070** (0.034)	−0.176*** (0.035)	−0.092** (0.041)	0.074 (0.067)	−0.048 (0.043)
<i>Net Borrowing (%)</i>		0.704*** (0.075)	0.467*** (0.031)	0.376*** (0.035)	0.524*** (0.037)	0.719*** (0.056)	−0.082 (0.073)
<i>Foreign Asset (%)</i>		−0.589 (0.522)	0.137 (0.137)	0.139 (0.127)	0.107 (0.155)	0.063 (0.177)	0.044 (0.313)
<i>KA Open</i>		−0.746*** (0.253)	−0.008 (0.214)	0.189 (0.205)	0.002 (0.246)	0.427 (0.506)	−0.235 (0.295)
<i>Δ Private Credit (%)</i>		−0.234*** (0.055)	−0.194*** (0.017)	−0.185*** (0.016)	−0.230*** (0.025)	−0.303*** (0.046)	−0.145*** (0.035)
<i>Δ Term of Trade</i>		0.105*** (0.038)	0.094*** (0.012)	0.061*** (0.013)	0.105*** (0.014)	0.084*** (0.020)	0.032** (0.014)
<i>Population (under 14, %)</i>		32.852*** (5.722)	25.700*** (6.973)	22.400*** (6.947)	35.155*** (8.605)	29.318** (14.287)	−50.296*** (14.655)
<i>Population (over 65, %)</i>		−7.151 (7.680)	15.419 (10.876)	18.005* (10.555)	44.023*** (16.531)	119.972*** (30.229)	14.179 (56.648)
<i>Trade Openness</i>		0.053*** (0.007)	0.042*** (0.008)	0.041*** (0.007)	0.023** (0.010)	−0.051*** (0.018)	−0.047** (0.021)
Year		−0.950*** (0.113)	−1.251*** (0.261)	−0.136 (0.185)	−0.236*** (0.061)	−0.537*** (0.138)	0.119 (0.073)
Year FE	✓	✓	✓	✓	✓	✓	✓
Country RE			✓	✓	✓	✓	✓
Num.Obs.	2829	1560	1560	1463	1243	551	294
R2 Marg.			0.447	0.436	0.347	0.341	0.252
R2 Cond.			0.882	0.880	0.848	0.938	0.900
BIC	46 158.5	22 594.1					

* p < 0.1, ** p < 0.05, *** p < 0.01

Table C.7: Regime Type's Effect on Trade Balance.

Since property-rights protection and rule of law have quite different distributions across

autocratic WTO-joiners, I make sure both low and high institutional ranges contain at least some autocracies that joined the WTO during 1990-2020. The separation looks like $\{0, 0.3, 0.7, 1\}$. For each range, I compare autocracies to all democracies that joined during the same period to keep the control group the same and I dichotomize polity into a democracy dummy so that the interaction effect (WTO x polity) doesn't reflect within-democracy variation.

PanelMatch (country-years)

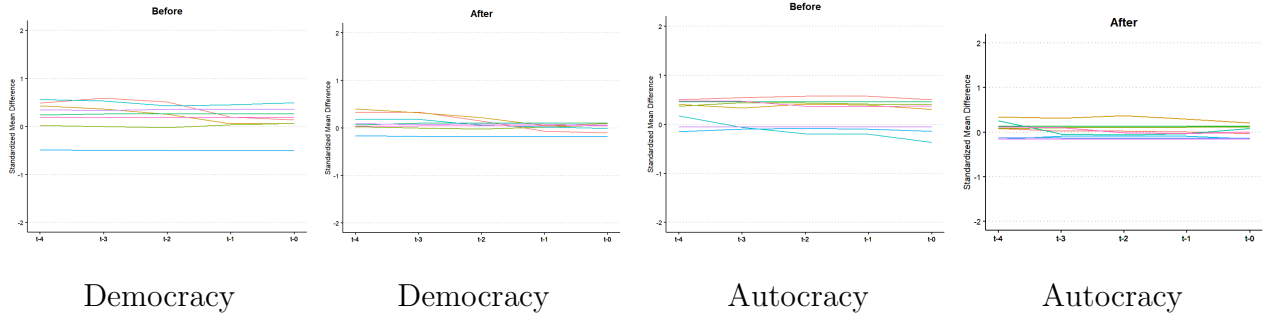


Figure C.12: Improved Covariate Balance via CBPS Weighting, Post-1990. *Note:* overall, all covariates' balance improve significantly. The green line natural resource intensity is slightly not balanced. However, it may not be an concern as it does not significantly affect WTO accession theoretically.

PanelMatch (dyad-years)

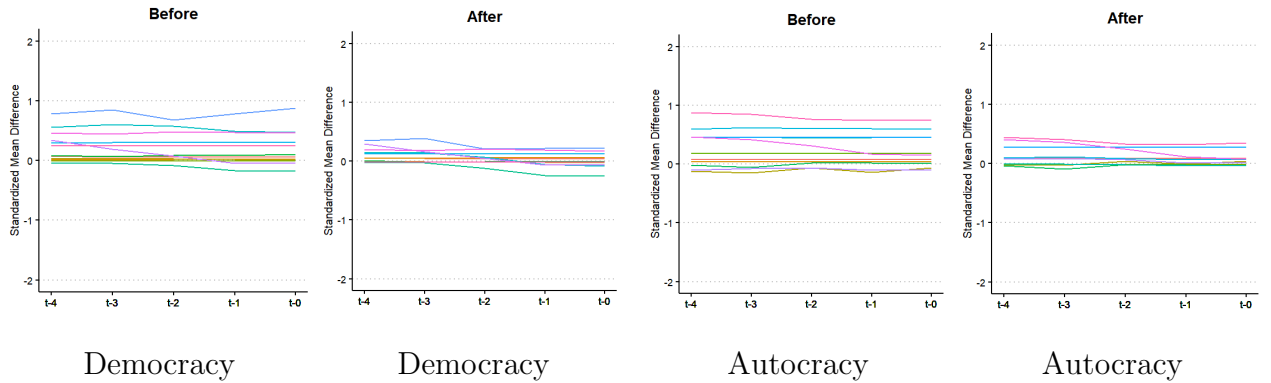
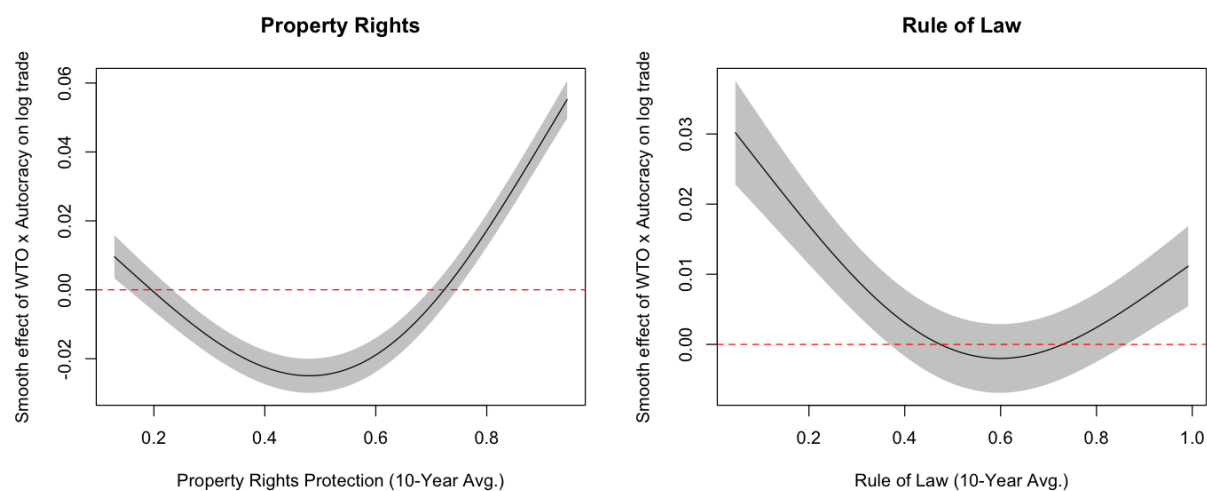


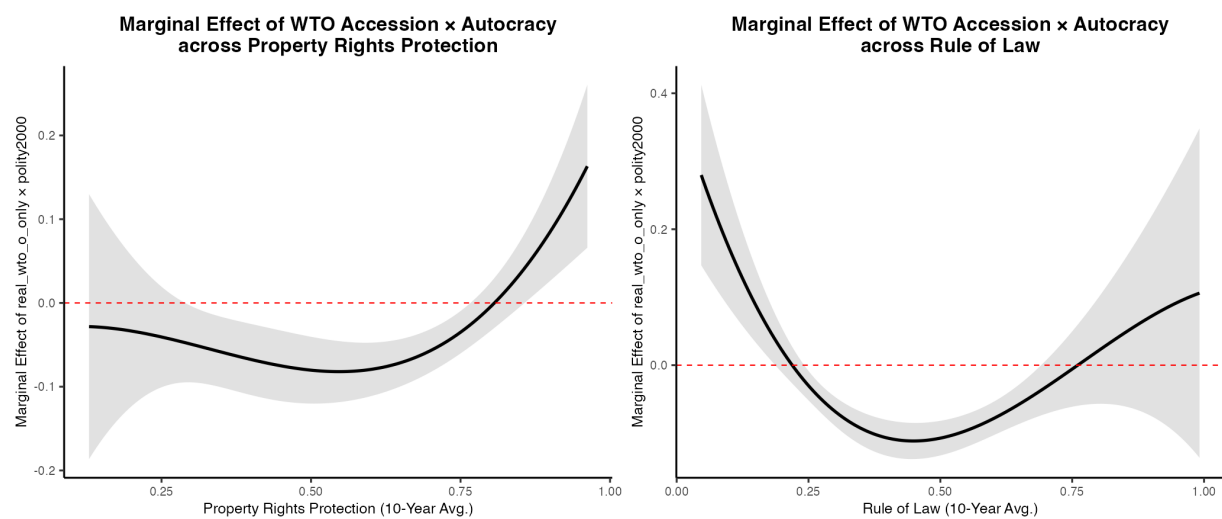
Figure C.13: Improved Covariate Balance via CBPS Weighting, Post-1990. *Note:* overall, all covariates' balance improve significantly.

C.4 Moderated WTO Effect across Institutional Range

General Additive Model (GAM)



a. GAM



b. Spline

Figure C.14: The U-Shape Moderated Effect of Polity across Institutional Ranges.

C.5 Domestic Reform Effect

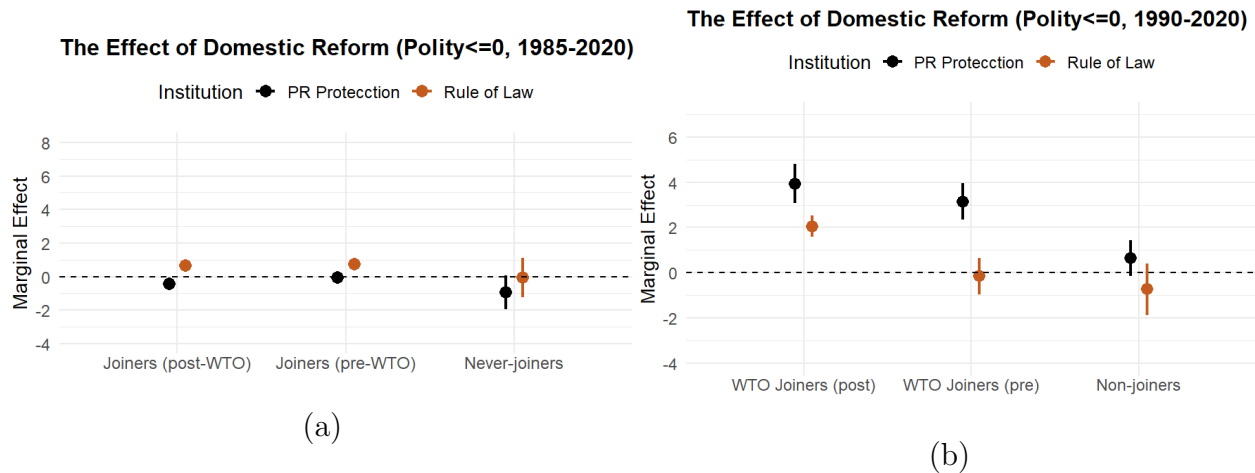


Figure C.15: The Effects of Domestic Reform by WTO-related Category. *Note:* (a) plots the effects of within-dyad changes of institutions by WTO-related Category for democracies only (Polity > 0 in 2000). I only include developing countries (GDP per capital lower than \$20,000 in 2000) to focus on institutional reform. (b) plots the effects of within-dyad changes of institutions for autocracies only. “Joiner” means a country joined the WTO during 1990-2020.

D Test using Engaged-Reformer Status

The tests in Section 5 classify institutional reform by a country’s average rule-of-law score over the ten years following accession. That measure is fixed within a country: it cannot distinguish years in which a country was genuinely unreformed from years in which it had already reformed, and it therefore assigns a country’s early, less-reformed years to whatever bin its eventual trajectory produces. Because the theory’s object of interest is the joint condition of engagement and reform, I construct that condition directly as a time-varying indicator.

Let $reform_{it} = 1$ if country i ’s V-Dem rule-of-law score in year t falls within the moderate band $[0.2, 0.7]$ —the same range used in the main analysis, derived from the 20th and 80th percentiles among developing countries (Appendix B.3). Because V-Dem scores fluctuate from year to year, I require the condition to hold in two consecutive years before coding a country as reformed, so that movements near the boundary do not toggle status. Engaged-

reformer status is then:

$$er_status_{it} = \mathbf{1}\{in_wto_{it} = 1 \wedge reform_confirmed_{it} = 1\}.$$

The indicator is time-varying and reversible: a country enters engaged-reformer status when it satisfies both conditions and exits if either lapses.

I estimate the effect using PanelMatch, the same matching-based approach applied to WTO accession in Section 5. For each country-year in which status switches from zero to one, the estimator constructs a matched set of control country-years from units with an identical treatment history over the preceding four years, then compares the treated unit’s change in log exports to the weighted change among its matched controls over the following five years. Matching covariates are GDP, GDP per capita, Polity, population, ethnic composition, NATO membership, natural-resource and industrial intensity, and lagged log exports. The rule-of-law score is excluded, since it defines part of the treatment; NATO membership is dropped from the autocracy-restricted samples, where it does not vary.

One departure from the main-text specification warrants note. Refinement uses Mahalanobis distance rather than covariate-balancing propensity score weighting: restricting the sample to a single regime type reduces the number of eligible controls per matched set enough that the CBPS weighting model is not identified for some treated units. Mahalanobis matching requires no model fit and is robust to small matched sets. Standard errors are bootstrapped, and placebo estimates for the four pre-treatment periods test the parallel-trends assumption. The sample runs 1986–2018, stratified into democracies and autocracies by Polity in 2000.

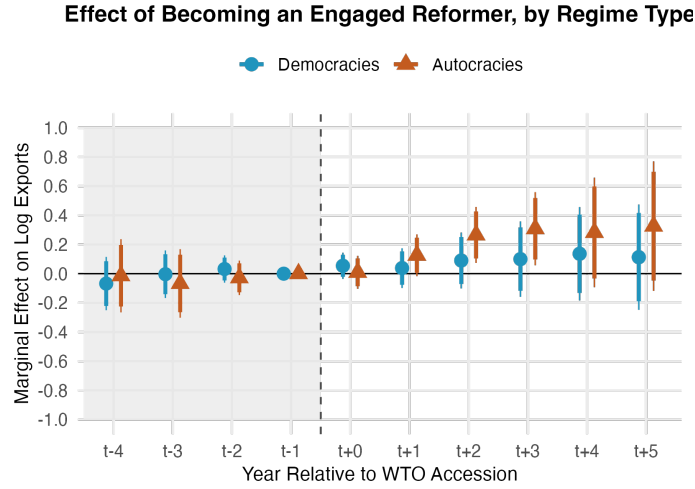


Figure D.16: Effect of Becoming an Engaged Reformer, by Regime Type. *Note:* Engaged-reformer status is a time-varying indicator equal to one when a country is both a WTO member and within the moderate rule-of-law band $[0.2, 0.7]$ for two consecutive years. Panel-Match with Mahalanobis refinement, four pre-treatment and five forward periods, 1986–2018. Shaded region denotes placebo estimates; $t - 1$ is the reference. Thick and thin bars denote 90% and 95% confidence intervals.

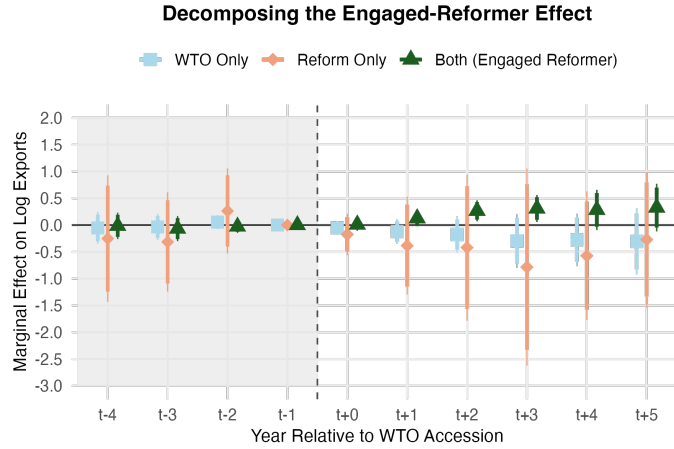


Figure D.17: Decomposing the Engaged-Reformer Effect. *Note:* Effects among autocracies of three treatments against a common baseline of countries satisfying neither condition. The “both” arm reproduces the autocracy estimate in Figure D.16. The isolated components are estimated from four and twelve countries and are correspondingly imprecise; note the wider vertical scale. Specification otherwise as above.

Figure D.16 reports estimates by regime type. Pre-treatment estimates are flat and statistically indistinguishable from zero for both groups, consistent with parallel trends. Following onset, autocracies’ exports rise steadily, reaching 0.26 log points at $t + 2$ and 0.31 at $t + 3$, with confidence intervals excluding zero; democracies show a smaller effect that remains indistin-

guishable from zero at every horizon. Intervals for autocracies widen at $t+4$ and $t+5$, so the effect is estimated most precisely in the first three years after onset. Figure D.17 decomposes engaged-reformer status among autocracies, comparing three treatments against a common baseline of countries satisfying neither condition: WTO membership without reform, reform without WTO membership, and both. Only the joint condition produces export gains; the “both” arm reproduces the autocracy estimate above and is shown again for comparison against the isolated components.

The isolated components are estimated from few countries—four for WTO-only and twelve for reform-only—and are correspondingly imprecise, with standard errors between 0.3 and 1.2. The reform-only arm also shows a positive pre-treatment estimate at $t-2$, so its parallel-trends assumption is questionable. These estimates should not be read as evidence that either condition alone reduces exports; the defensible inference is that neither component alone reproduces the gain associated with their conjunction. That few autocracies sustain either condition without the other is itself consistent with the scope condition operating as a joint threshold.

E Mechanism Tests

F Robust Tests

F.1 The Domestic Reform Effect

I now test whether or not domestic reform favors autocracies. I also use gravity models to test the effect of domestic reform on trade. VDem’s PR protection and rule of law are used to measure institutional levels. I exploit within-dyad variation with dyad and year fixed-effects which control for possible time- and dyad-invariant confounders. As reform is mostly for developing countries, I focus on those with GDP per capita below \$20,000 in 2000, which results in 165 countries. For the case of post-1990 trade expansion, I also test the case of outsiders to align with my theory on post-1990 globalization: new-joiners (who joined the

WTO after 1990) and never-joiners – 94 total with 30 autocracies. I assign countries into four ranges based on Polity score in 2000 as Polity is quite stable for post-1990 period. Institutions are lagged by one year.

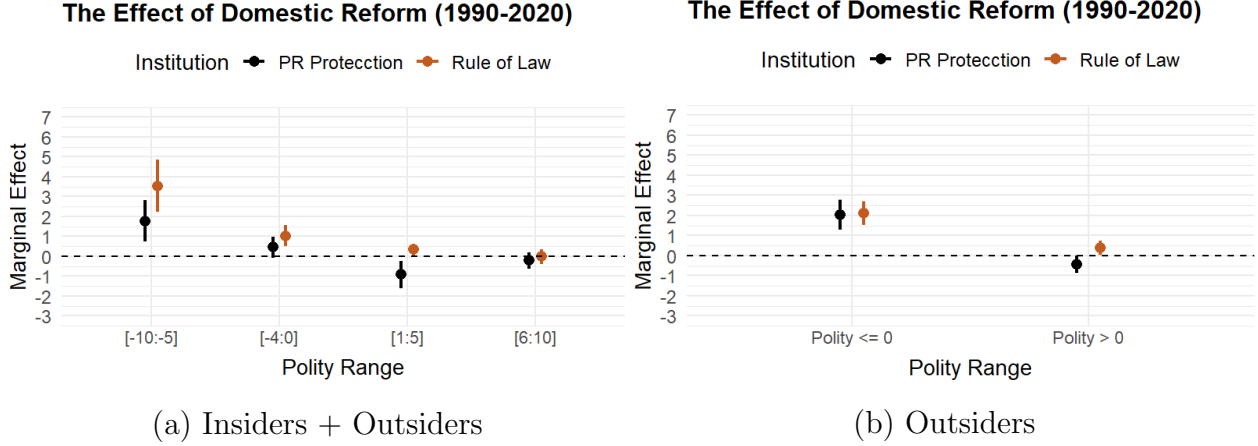


Figure F.18: The Effects of Domestic Reform by Polity. *Note:* (a) plots the effects of within-dyad institutional changes across ranges of Polity, for all developing countries (GDP per capital below \$20,000 in 2000). (b) plots the same graph, but only for outsiders (new-joiners and never-joiners). Results are consistent for including both institutional measures.

As shown in Figure F.18a, the effects of domestic reform among developing countries during the period of 1990-2020 are generally higher for more autocratic states.⁵³ This pattern is more apparent in Figure F.18b, where I remove those who already joined the WTO before 1990. The likely interpretation is that the influx of many well-performing autocracies into the global trade system may exert significant stress to more hands-off, open-market democracies, with an exemplary case being the China shock. The result is consistent with the hypothesis that autocratic advantages amplify the effect of domestic reform. Admittedly, autocracies may also have increased marginal returns due to lower starting institutional levels. However, developing democracies' institutions were not significantly higher: 0.6 vs. 0.45 (autocracy) for PR protection in 1990 (see Figure 5), which cannot explain why democracy's reform effect is near zero.

Trade Integration as the Scope Condition

⁵³I don't control for WTO membership to avoid the post-treatment bias. Similar effects remain with WTO membership.

As the effect of domestic reform is only substantially positive for autocratic outsiders (Figure F.18b), I focus only on the dyad-years in which exporter is an autocracy in 2000.⁵⁴ Assuming the main gravity model that controls for trade’s determinants is still valid, I fit a model including an interaction term to estimate effects for two strata: joiners (post-1990 pre-WTO period) and joiners (post-WTO period), for both autocracy and democracy, respectively:

$$Export_{ijt} = \beta Institution_{i,t-1} \times Polity_i \times WTO_{it} + \delta \mathbf{X}_{ijt} + \gamma_{ij} + \eta_t + \epsilon_{ijt}$$

where $\mathbf{X}_{ijt}$ denotes dyad-level covariates. $Institution_{i,t-1} \times WTO_{it} \times Polity_i$ captures the effect of institution (lagged) moderated by WTO period classified by Polity in 2000. $\mathbf{X}_{ijt}$ denotes dyad-level covariates. I include dyad and year fixed-effects. Additionally, I estimate the non-interaction version of the model for joiners’ pre-1990 reform period (1975-1989) before they were substantively engaged, and never-joiners.

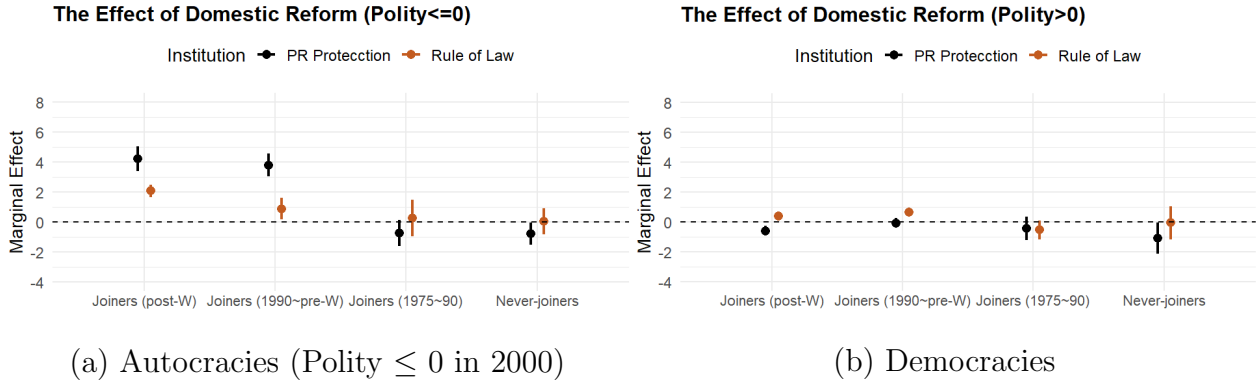


Figure F.19: The Effects of Domestic Reform by Period. *Note:* Joiners are those joining the WTO after 1990. “W” denotes the WTO.

Figure F.19a displays the effects of domestic reform for four categories. For PR protection, every 0.1 increase leads to 65.3% in exports for joiners’ pre-WTO period, 54.3% change for joiners’ post-WTO periods, and almost no change for joiners’ pre-1990 period (1975-1989) and never-joiners. For rule of law, the effects are 22.2%, 13.6%, null, and null, respectively.

⁵⁴I include three years earlier (i.e., 1987-89) to allow for pre-wto years for those joined in the early 1990s, though no inclusion doesn’t affect results.

By comparison, democracies see almost null effects for all categories (Figure F.19b). Note that for joiners in the pre-WTO period (post-1990), as mentioned, many already enjoyed globalization benefits through semi-engagement, such as MFNs and PTAs, RTAs, spill-over from joiners, and future WTO prospect, so we expect positive effects compared to the non-engagement counterfactual.⁵⁵ Meanwhile, joiners' post-WTO period tend to have higher institutional level than pre-WTO period, which may have diminishing marginal returns of reforms, making the observed effect lower than that with a similar baseline as the pre-WTO period. In other words, WTO period may lead to higher effect than observed. Finally, for never-joiners, their effects are always null, although they may have other systemic differences. Overall, the results support my theory – autocratic advantages manifest conditional on the engagement by the global trade regime.

G Qualitative Tests

Belarus vs. Ukraine - Belarus and Ukraine shared many similarities in the early 1990s: per capita income, political legacies, geography, resource endowments, and cultural and racial back- grounds. Both countries implemented market-oriented reforms to a similar level (0.75 vs. 0.9 in PR protection, and 0.26 vs. 0.2 in rule of law in 2000). While Belarus never joined the WTO, it was semi-engaged through channels such as the MFNs (with the U.S. and EU in the 1990s), economic spill-over from Russia, the China-Belarus industrial park, and RTAs like the EAEU (Eder 2021). Ukraine, by contrast, joined the WTO in 2008. Politically they diverged: Belarus is autocratic and Ukraine practices democracy. It turns out Belarus's centralized, state-led gradual reforms more favored industrial development (Ibid), whereas Ukraine's fragmented political system subject to interest groups, oligarchs, and "hands-off" liberalization has left it vulnerable to external shocks and underinvestment in infrastructure and industries (Kuzio 2020). Between the early 1990s and the mid-2010s, Belarus increased its exports thirteen-fold, significantly outpacing Ukraine, which saw only a five-fold increase alongside a per capita income half of Belarus'.

⁵⁵I don't use lead WTO dummies as two periods still differ substantively.

The UAE vs. Kuwait – While neither country is a democracy, it nonetheless pertains to my theory. Although the UAE and Kuwait share comparable resource endowments (holding 5.9% and 6.1% of global oil reserves, respectively), as well as similarities in GDP per capita during the 1990s, geographic location, population size, religion, and ethnic composition, notable institutional differences help explain their divergent performance trajectories. The UAE’s federal monarchy, with its highly centralized authority across seven emirates, has led to government-led development, fast decision-making, and a flexible labor regime, enabling swift, coordinated economic policies and large-scale investments in infrastructure such as extensive transport networks which have fostered a diversified export base (Hvidt 2013). In contrast, Kuwait’s constitutional emirate, with a relatively empowered parliament, often exhibits more fragmented governance and slower policy reactions and suffers from “the lack of political consensus and long-overdue reforms” (Ibid), despite its accession into the WTO (1963) much earlier than the UAE (1994).

Russia vs. Brazil - Russia and Brazil also shared some similarities: similar income levels in the early 1990s, regional resource-oriented powers with similarly vast landmass and population, and similar economic problems and market reforms in the late 1980s. In Russia’s case, the implementation of “shock therapy,” followed by a series of stabilization efforts and partial re-nationalization of key industries, was deeply facilitated by its centralized state and authoritarian legacy (Aslund 2012). Russia, along with many other post-Soviet states, strategically devalued its currency to stimulate exports, maintained low taxes, limited social transfers, and suppressed wage growth (Ibid). These partly explain Russia’s faster growth rate than most Central European countries that adopted the EU model, reinforcing the “old idea of authoritarian advantage” (Ibid). Consequently, Russia’s exports surged in the 1990s and remained at a high level afterwards (Figure 19), while keeping current account surpluses for decades. In contrast to Russia’s late WTO accession (2012), Brazil was a GATT signatory. While Brazil’s energy sector is more diverse than Russia’s, it also heavily relies on resource exports (World Bank). As a democracy, Brazil experienced recurrent policy instability, driven by electoral turnover and fiscal pressures to expand social spending, which in turn

created an uncertain business climate and obstructed long-term structural reforms (Franko 2018). Brazil's trade policies which merely shield domestic interest groups and stricter labor regulations constrained competitiveness and innovation (Ibid; Feierherd 2024). Although Brazil enjoyed a temporary export growth during the 2000s commodity boom, its overall exports has remained substantially lower than Russia's (Figure 19), with consistently recorded current account deficits for decades.

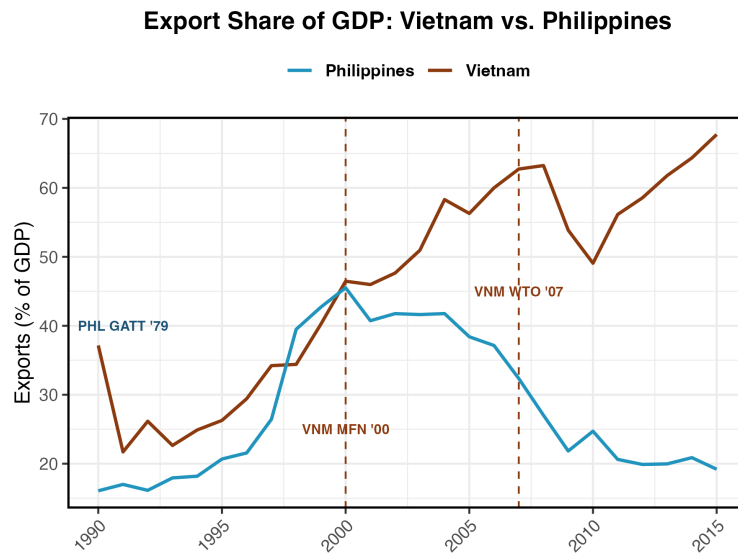


Figure G.20: Exports (%): Vietnam and the Philippines.